Exact Inference

Variable Elimination

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Module 3 introduces a key algorithm for Exact Inference known as Variable Elimination (VE; Chapter 9).

We are going to use VE in two kinds of inference problems: sum-product inference (Chapter 9) and max-product inference (Chapter 13).

ILOs After this module the student

- can perform sum-product and max-product inference via VE;
- can predict the complexity of VE by constructing an induced graph;
- can optimise VE by ordering elimination steps opportunistically.

Textbook for this course: Koller and Friedman [1].

Overview of Module 3

HC3a: naive sum-product inference and variable elimination.

LC3: VE in code.

HC3b: elimination ordering and max-product inference.

WC3: exam-like exercises.

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Naive Exact Inference

Conditional Probability Queries [section 2.1.5.1]

We have a joint distribution $P(\mathbf{X})$ over a collection of rvs $\mathbf{X}$, this distribution is represented by a PGM such as a BN or MN.

Conditional probability queries:

- **Evidence:** an assignment $\mathbf{E} = \mathbf{e}$ of a subset $\mathbf{E} \subseteq \mathbf{X}$ of the rvs.
- **Query:** given the evidence, we are interest in reasoning about the possible assignments of the rvs in $\mathbf{Q} \subseteq \mathbf{X}$, where $\mathbf{Q} \cap \mathbf{E} = \emptyset$.
- **Task:** express $P(\mathbf{Q}|\mathbf{E} = \mathbf{e})$

Applications: medical/fault diagnosis, image/text completion, speech analysis, ...

Conditional Probability Queries [section 2.1.5.1]

$$P(\mathbf{Q}|\mathbf{E}=\mathbf{e}) = \frac{P(\mathbf{Q},\mathbf{E}=\mathbf{e})}{P(\mathbf{E}=\mathbf{e})} \propto P(\mathbf{Q},\mathbf{E}=\mathbf{e})$$

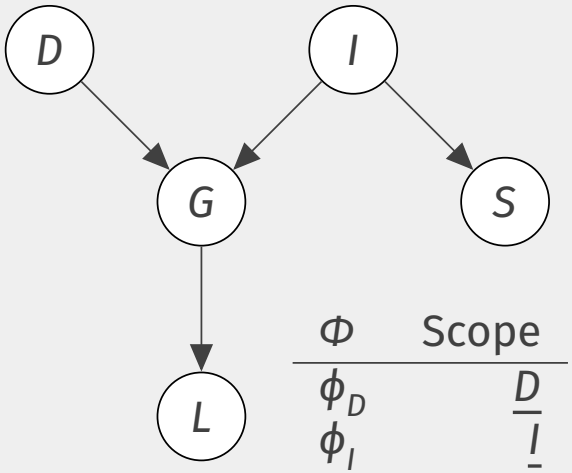
Let $\mathbf{W}$ be the complement of $\mathbf{E}$ and $\mathbf{Q}$ in $\mathbf{X}$. That is, the 'rest' of the variables in the model.

Then, we can express $P(\mathbf{Q},\mathbf{E}=\mathbf{e})$ via marginalisation of $\mathbf{W}$. This is an instance of what we call inference. Specifically, *marginal inference*:

$$P(\mathbf{Q},\mathbf{E}=\mathbf{e}) = \sum_{\mathbf{W}} P(\mathbf{W},\mathbf{Q},\mathbf{E}=\mathbf{e})$$

To continue, we need to know more about how $P(\underbrace{\mathbf{W},\mathbf{Q},\mathbf{E}}_{\mathbf{X}})$ factorises.

A factor-view of the *Student* example



Φ	Scope
ϕ_D	$\underline{D}$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$

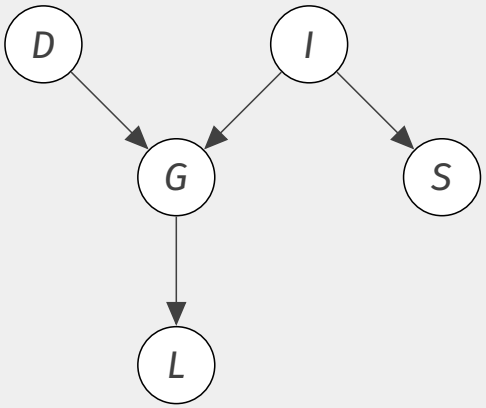
In factor form, we use ϕ_X with scope $\{X\} \cup \text{Pa}(X)$ for $P(X|\text{Pa}(X))$.

Joint distribution:

$$\begin{aligned} P(D, I, S, G, L) &= P(D)P(I)P(S|I)P(G|D, I)P(L|G) \\ &= \phi_D(D)\phi_I(I)\phi_S(S, I)\phi_G(G, D, I)\phi_L(L, G) \end{aligned}$$

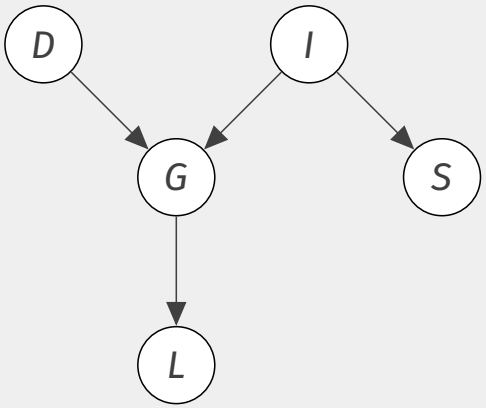
In this view, we can develop a framework for inference that works both for BNs and MNs.

Exact Inference: Sum-Product *Student* example



Evidence: $D = d^1$. Query: L.

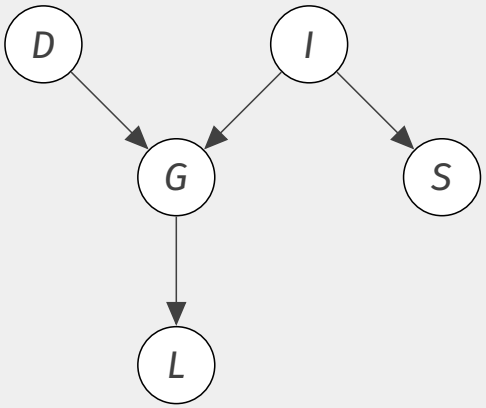
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To express $P(L|D = d^1)$

1. assign $D = d^1$:
   $P(D = d^1, I, S, G, L)$

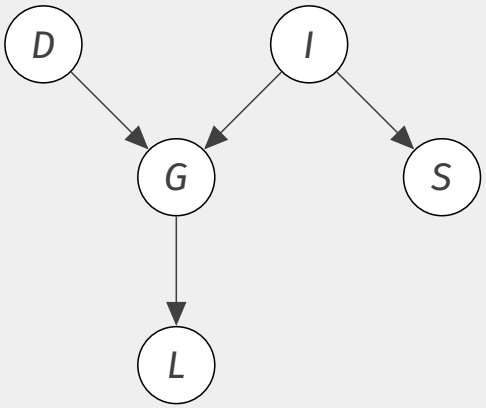
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2. marginalise I, G, S:
   $P(L, D = d^1)$;

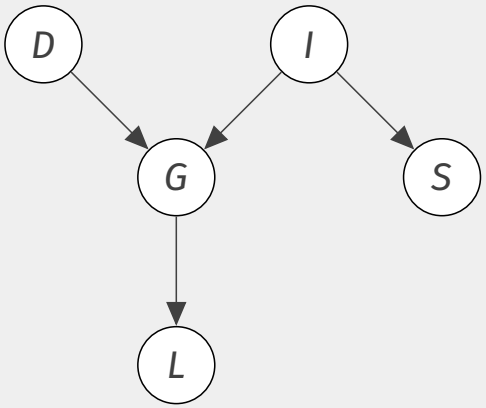
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3. normalize over Val(L).

Exact Inference: Sum-Product *Student* example



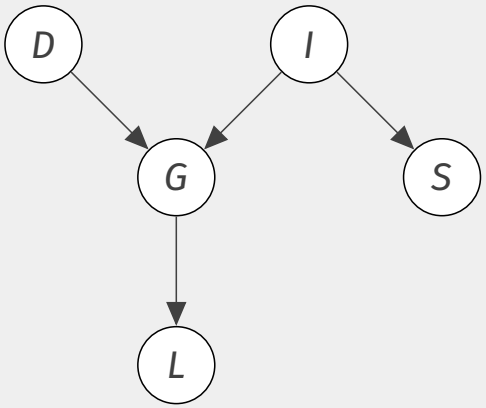
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Start with the joint distribution, but let's write it using *factors*:

$$P(D, I, G, S, L)$$
$$= \phi_D(D)\phi_I(I)\phi_G(G, D, I)\phi_S(S, I)\phi_L(L, G)$$

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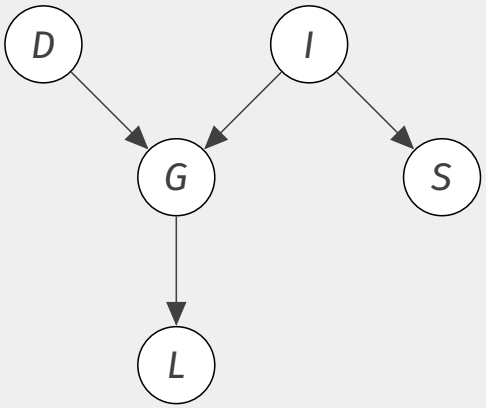
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1. Reduce the factors to the available evidence:

$$P(D = d^1, I, G, S, L)$$
$$= \phi_D[d^1]\phi_I(I)\phi_G[d^1](G, I)\phi_S(S, I)\phi_L(L, G)$$

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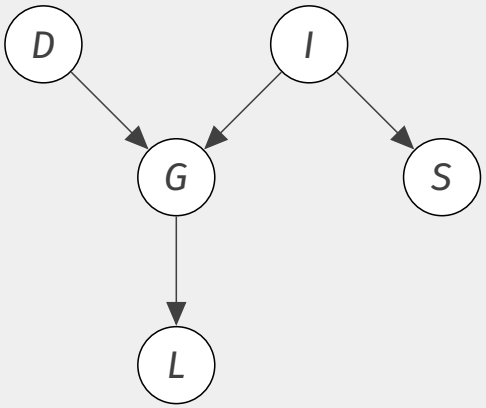
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2. Marginalise the variables that are neither assigned nor queried about:

$$P(L, D = d^1)$$
$$= \sum_{G,I,S} \phi_D[d^1]\phi_I(I)\phi_G[d^1](G, I)\phi_S(S, I)\phi_L(L, G)$$

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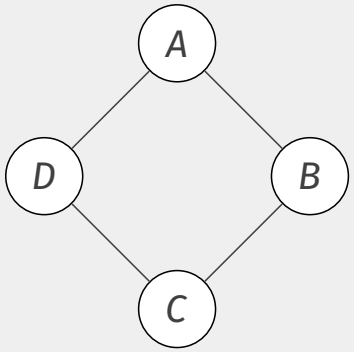
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3. Normaliser: $\sum_{l \in \text{Val}(L)} P(L = l, D = d^1)$

Exact Inference: Sum-Product *Misconception* example

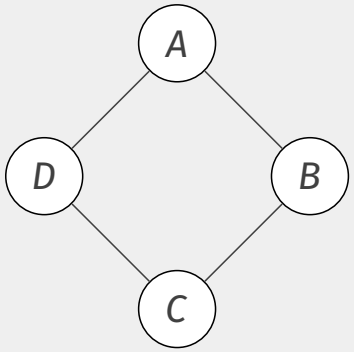


Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

Evidence: $A = a^1, C = c^1$.
Query: B.
To express $P_\Phi(B|A = a^1, C = c^1)$

1. assign $A = a^1, C = c^1$:
   $\tilde{P}_\Phi(A = a^1, B, C = c^1, D)$

Exact Inference: Sum-Product *Misconception* example

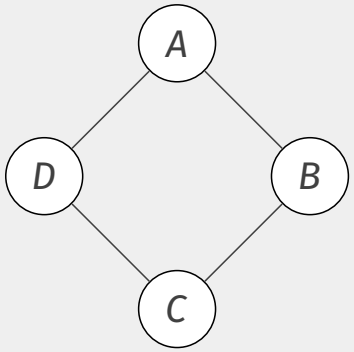


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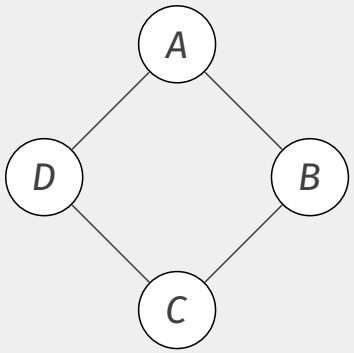


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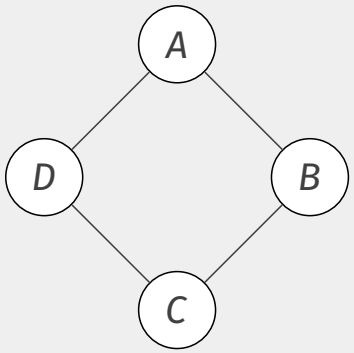
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Start with the unnormalised measure:

$$\tilde{P}_\Phi(A, B, C, D)$$
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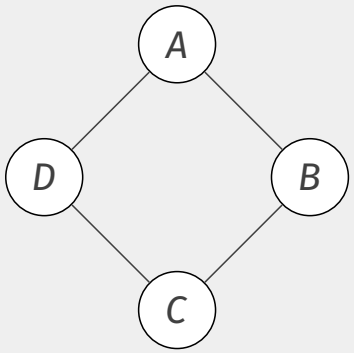
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$$\tilde{P}_\Phi(A, B, C, D) = \phi_1(A, B)\phi_2(B, C)\phi_3(C, D)\phi_4(D, A)$$

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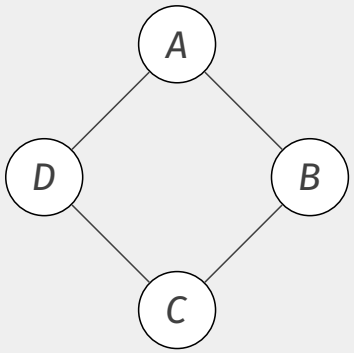
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2. Marginalise the variables that are neither assigned nor queried about:

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3. Normaliser:
$\sum_{b \in \text{Val}(B)} \tilde{P}_\Phi(B = b, A = a^1, C = c^1)$

Exact Inference

Naive marginal inference requires creating the full joint (unnormalised) distribution—a large product—before we can marginalise (sum). Hence the name 'sum-product'.

Of course, we can reduce factors to the evidence before taking the large product, but you can see that this is not an efficient algorithm.

No matter the graphical structure, it will always involve a product of all factors, which, in the worst case, is a very large factor.

Next, we use the graphical structure to obtain better average performance via the so-called variable elimination algorithm.

Variable Elimination

Exploiting the Graph Structure

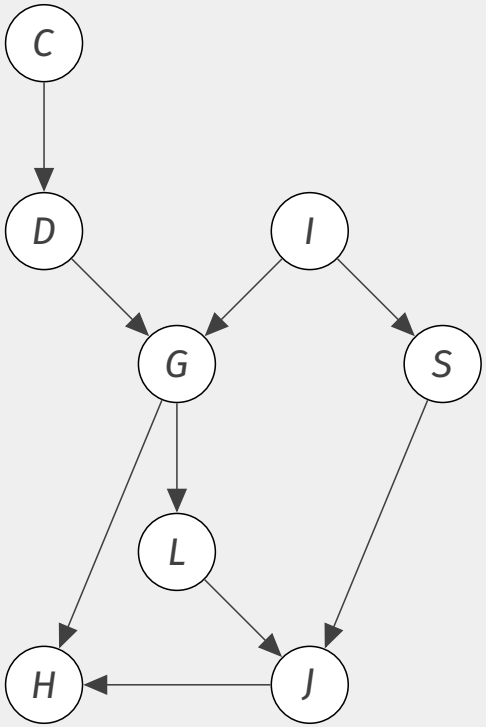
Naive sum-product does not use the graph structure at all. It starts from a full product of all factors.

The key challenge in inference is marginalisation of unassigned rvs. We build the entire product of factors because then we are sure that it is safe to marginalise rvs.

But consider marginalising one rv at a time, by looking around in the graphical structure, we might be able to see that not all factors are going to be relevant.

This intuition is the key to a more 'incremental' procedure known as variable elimination (VE; Section 9.3).

Extended Student (Fig 9.8)

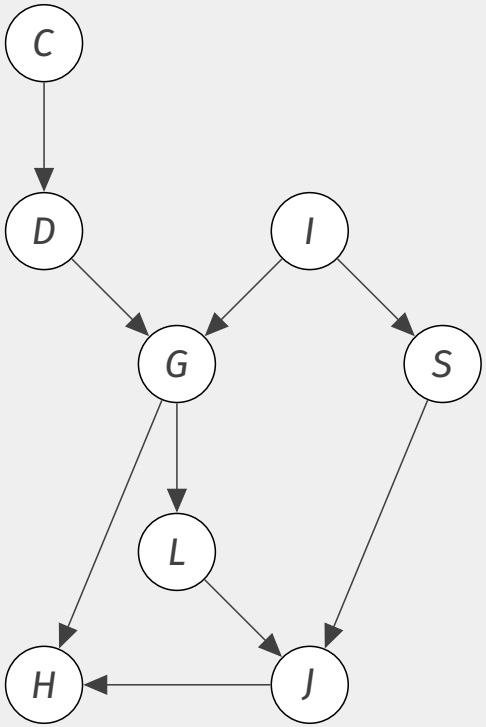


In factor form, we use ϕ_X with scope $\{X\} \cup \text{Pa}(X)$ for $P(X|\text{Pa}(X))$.

Joint distribution:

$$P(C, D, I, G, S, L, JH) = \phi_C(C)\phi_D(D, C)\phi_I(I)\phi_S(S, I) \times \phi_G(G, D, I)\phi_L(L, G)\phi_J(J, L, S)\phi_H(H, G, J)$$

Extended Student (Fig 9.8)

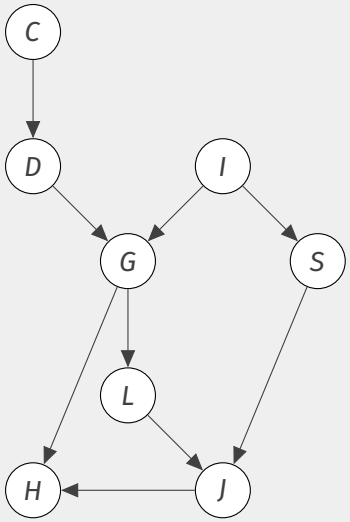


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Example: let's express $P(J)$ by eliminating C, D, I, H, G, S, L in this order [example 9.1]



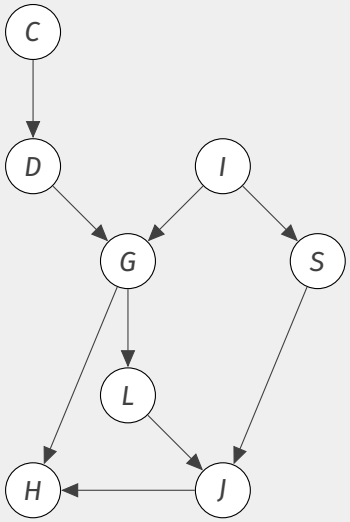
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Goal: $P(J)$.

Eliminate: $\underline{C}, D, I, H, G, S, L$.

$$\sum_C \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I)\phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$



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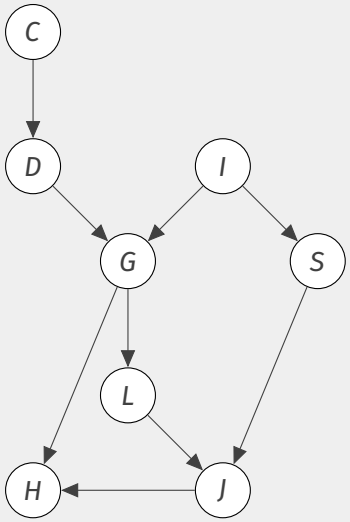
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Eliminate: $\underline{C}, D, I, H, G, S, L$.

$$\sum_C \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I)\phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

$$= \phi_I(I)\phi_S(S,I)\phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J) \underbrace{\sum_C \overbrace{\phi_C(C)\phi_D(D,C)}^{\rho_1(C,D)}}_{\tau_1(D)}$$



Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

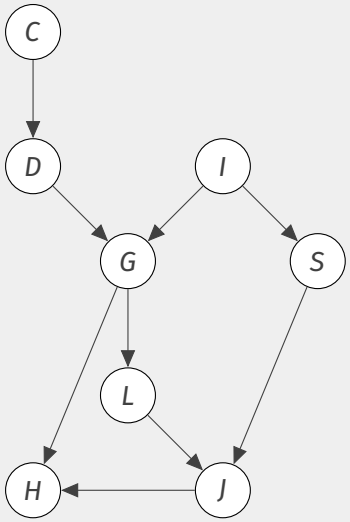
Goal: $P(J)$.

Eliminate: $\underline{C}, D, I, H, G, S, L$.

$$\sum_C \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I)\phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

$$= \phi_I(I)\phi_S(S,I)\phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J) \underbrace{\sum_C \overbrace{\phi_C(C)\phi_D(D,C)}^{\rho_1(C,D)}}_{\tau_1(D)}$$

$$= \phi_I(I)\phi_S(S,I)\phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_1(D)$$



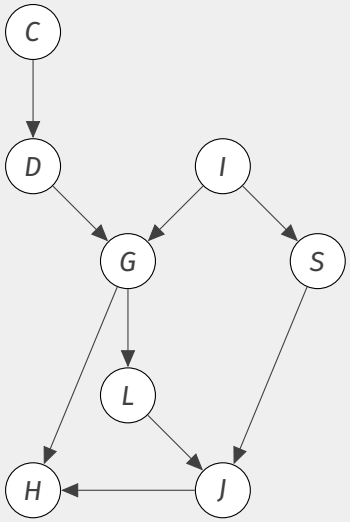
Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

Eliminate: $\not{C}, \underline{D}, I, H, G, S, L$.

$$\sum_D \phi_I(I)\phi_S(S,I)\phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_1(D)$$



Joint distribution:

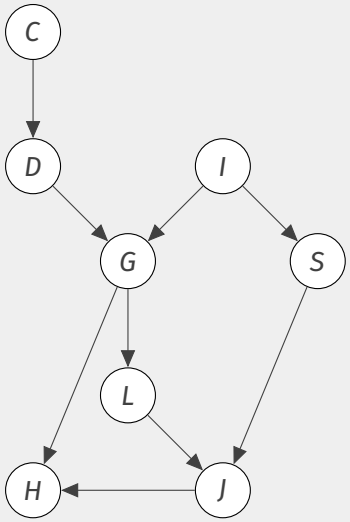
$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

Eliminate: $\not{C}, \underline{D}, I, H, G, S, L$.

$$\sum_D \phi_I(I)\phi_S(S,I)\phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_1(D)$$

$$= \phi_I(I)\phi_S(S,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J) \underbrace{\sum_D \overbrace{\tau_1(D)\phi_G(G,D,I)}^{\rho_2(G,D,I)}}_{\tau_2(G,I)}$$



Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

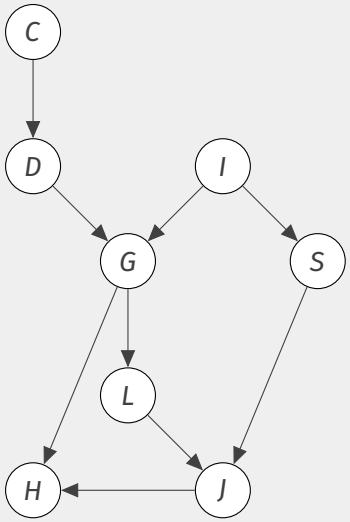
Goal: $P(J)$.

Eliminate: $\not{C}, \underline{D}, I, H, G, S, L$.

$$\sum_D \phi_I(I)\phi_S(S,I)\phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_1(D)$$

$$= \phi_I(I)\phi_S(S,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J) \underbrace{\sum_D \overbrace{\tau_1(D)\phi_G(G,D,I)}^{\rho_2(G,D,I)}}_{\tau_2(G,I)}$$

$$= \phi_I(I)\phi_S(S,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_2(G,I)$$



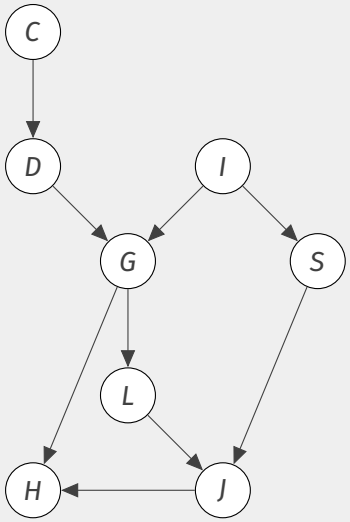
Eliminate: $\not{C}, \not{D}, \underline{I}, H, G, S, L$.

Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_I \phi_I(I)\phi_S(S,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_2(G,I)$$



Joint distribution:

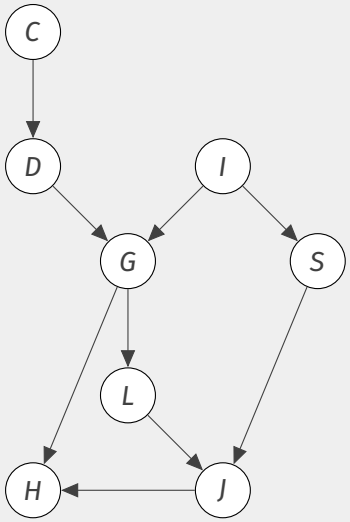
$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

Eliminate: $\not{C}, \not{D}, \underline{I}, H, G, S, L$.

$$\sum_I \phi_I(I)\phi_S(S,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_2(G,I)$$

$$= \phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J) \underbrace{\sum_I \overbrace{\phi_I(I)\phi_S(S,I)\tau_2(G,I)}^{\rho_3(I,G,S)}}_{\tau_3(G,S)}$$



Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

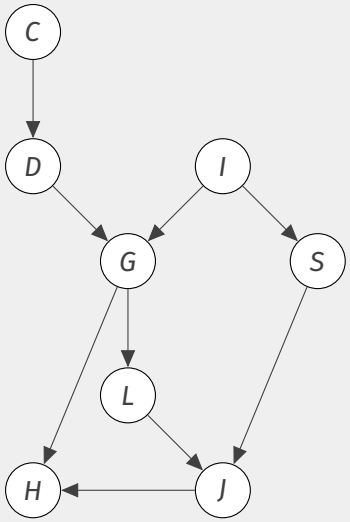
Goal: $P(J)$.

Eliminate: $\not{C}, \not{D}, \underline{I}, H, G, S, L$.

$$\sum_I \phi_I(I)\phi_S(S,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_2(G,I)$$

$$= \phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J) \underbrace{\sum_I \overbrace{\phi_I(I)\phi_S(S,I)\tau_2(G,I)}^{\rho_3(I,G,S)}}_{\tau_3(G,S)}$$

$$= \phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_3(G,S)$$



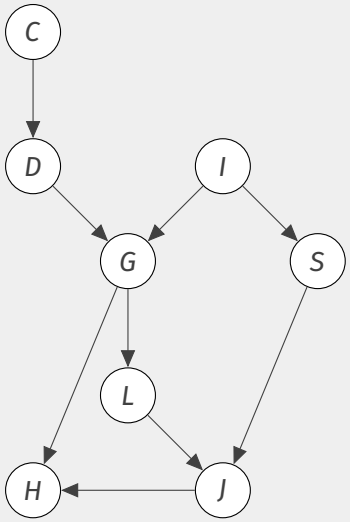
Eliminate: $\not{C}, \not{D}, \not{I}, \underline{H}, G, S, L$.

Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_H \phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_3(G,S)$$



Eliminate: $\not{C}, \not{D}, \not{I}, \underline{H}, G, S, L$.

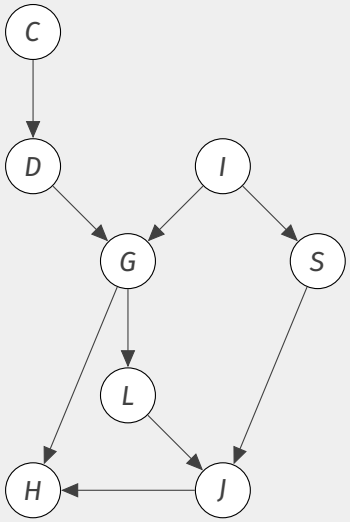
Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_H \phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_3(G,S)$$

$$= \phi_L(L,G)\phi_J(J,L,S)\tau_3(G,S) \underbrace{\sum_H \phi_H(H,G,J)}_{\tau_4(G,J)}$$



Eliminate: $\not{C}, \not{D}, \not{I}, \underline{H}, G, S, L$.

Joint distribution:

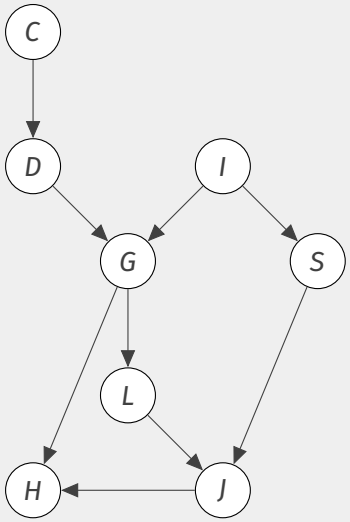
$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_H \phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_3(G,S)$$

$$= \phi_L(L,G)\phi_J(J,L,S)\tau_3(G,S) \underbrace{\sum_H \phi_H(H,G,J)}_{\tau_4(G,J)}$$

$$= \phi_L(L,G)\phi_J(J,L,S)\tau_3(G,S)\tau_4(G,J)$$



Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

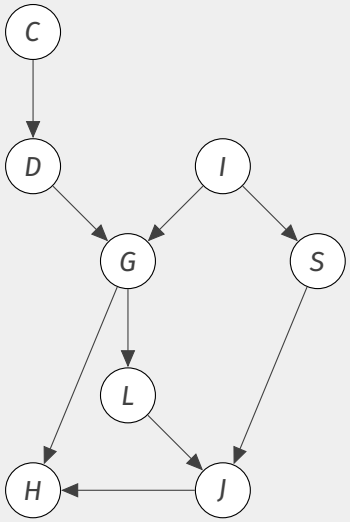
Eliminate: $\not{C}, \not{D}, \not{I}, \underline{H}, G, S, L$.

$$\sum_H \phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)\tau_3(G,S)$$

$$= \phi_L(L,G)\phi_J(J,L,S)\tau_3(G,S) \underbrace{\sum_H \phi_H(H,G,J)}_{\tau_4(G,J)}$$

$$= \phi_L(L,G)\phi_J(J,L,S)\tau_3(G,S)\tau_4(G,J)$$

$\tau_4(G,J) = 1$ but, for generality, let's pretend we do not know that.



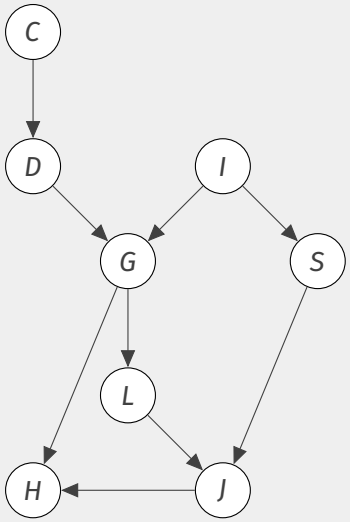
Eliminate: $\not{C}, \not{D}, \not{I}, \not{H}, \underline{G}, S, L$.

Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_G \phi_L(L,G)\phi_J(J,L,S)\tau_3(G,S)\tau_4(G,J)$$



Eliminate: ~~C~~, ~~D~~, ~~I~~, ~~H~~, $\underline{G}$, S, L.

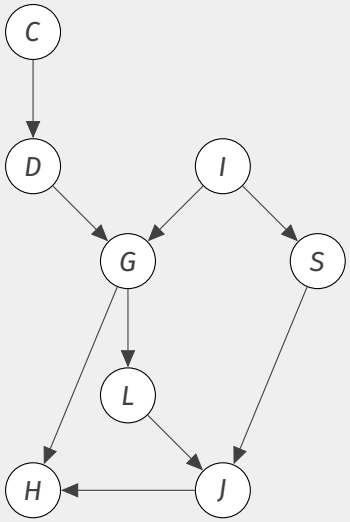
Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_G \phi_L(L,G)\phi_J(J,L,S)\tau_3(G,S)\tau_4(G,J)$$

$$= \phi_J(J,L,S) \underbrace{\sum_G \overbrace{\phi_L(L,G)\tau_3(G,S)\tau_4(G,J)}^{\rho_5(G,J,L,S)}}_{\tau_5(J,L,S)}$$



Eliminate: ~~C~~, ~~D~~, ~~I~~, ~~H~~, $\underline{G}$, S, L.

Joint distribution:

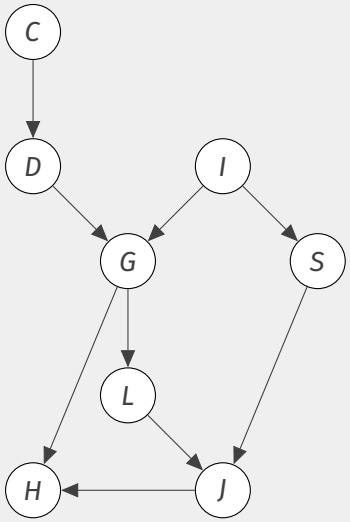
$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_G \phi_L(L,G)\phi_J(J,L,S)\tau_3(G,S)\tau_4(G,J)$$

$$= \phi_J(J,L,S) \underbrace{\sum_G \overbrace{\phi_L(L,G)\tau_3(G,S)\tau_4(G,J)}^{\rho_5(G,J,L,S)}}_{\tau_5(J,L,S)}$$

$$= \phi_J(J,L,S)\tau_5(J,L,S)$$



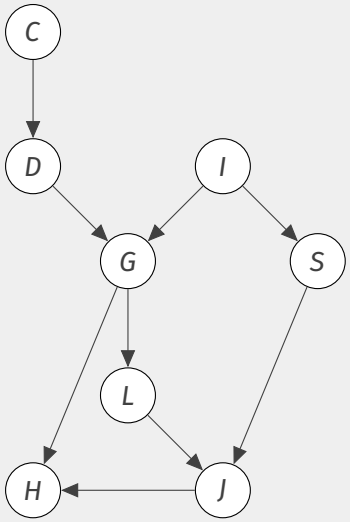
Eliminate: $\not{C}, \not{D}, \not{I}, \not{H}, \not{G}, \underline{S}, L$.

Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_S \phi_J(J,L,S)\tau_5(J,L,S)$$



Eliminate: ~~C~~, ~~D~~, ~~I~~, ~~H~~, ~~G~~, $\underline{S}$, L.

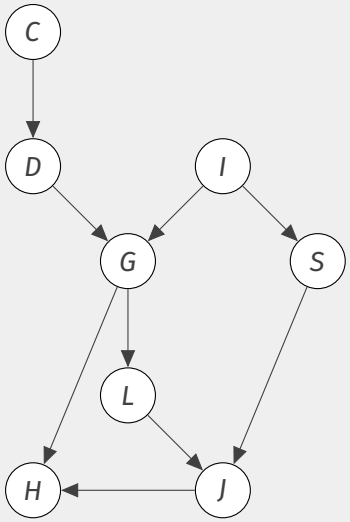
Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_S \phi_J(J,L,S)\tau_5(J,L,S)$$

$$= \underbrace{\sum_S \overbrace{\phi_J(J,L)\tau_5(J,L,S)}^{\rho_6(J,L,S)}}_{\tau_6(J,L)}$$



Eliminate: ~~C~~, ~~D~~, ~~I~~, ~~H~~, ~~G~~, $\underline{S}$, L.

Joint distribution:

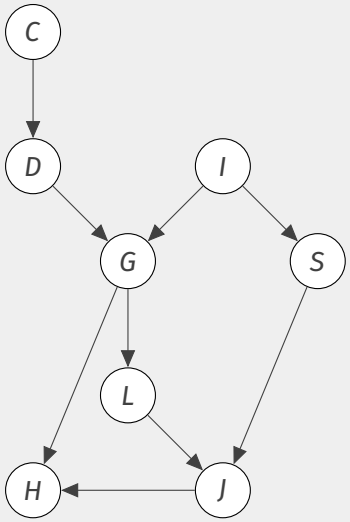
$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_S \phi_J(J,L,S)\tau_5(J,L,S)$$

$$= \underbrace{\sum_S \overbrace{\phi_J(J,L)\tau_5(J,L,S)}^{\rho_6(J,L,S)}}_{\tau_6(J,L)}$$

$$= \tau_6(J,L)$$



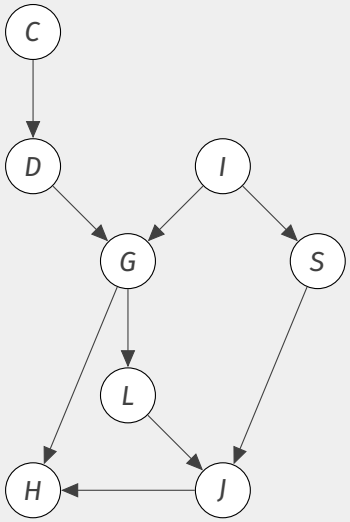
Eliminate: ~~C~~, ~~D~~, ~~I~~, ~~H~~, ~~G~~, ~~S~~, L.

Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_L \tau_6(J,L)$$



Eliminate: ~~C~~, ~~D~~, ~~I~~, ~~H~~, ~~G~~, ~~S~~, L.

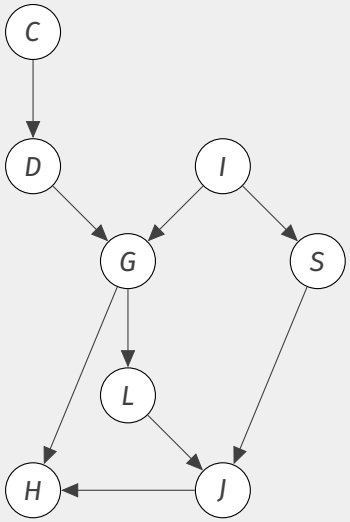
Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_L \tau_6(J,L)$$

$$= \tau_7(J)$$



Eliminate: ~~C~~, ~~D~~, ~~I~~, ~~H~~, ~~G~~, ~~S~~, L.

$\tau_7(J)$ is precisely $P(J)$.

Joint distribution:

$$P(C,D,I,G,S,L,JH) = \phi_C(C)\phi_D(D,C)\phi_I(I)\phi_S(S,I) \times \phi_G(G,D,I)\phi_L(L,G)\phi_J(J,L,S)\phi_H(H,G,J)$$

Goal: $P(J)$.

$$\sum_L \tau_6(J,L)$$

$$= \tau_7(J)$$

VE – Overview of Steps

We have a collection of factors Φ.
(They may be reduced to account for some evidence.)

For each rv Y that we want to marginalise, we

1. gather all relevant factors—those for which Y is in the scope—removing them from Φ;
2. take the product of the relevant factors;
3. marginalise Y out of the product;
4. add the resulting factor in the collection.

For example, marginalising G after having marginalised C, D, I, H

$$\sum_G \underbrace{\phi_L(L,G)\phi_J(J,L,S)\tau_3(G,S)\tau_4(G,J)}_{\text{(1) relevant}} = \phi_J(J,L,S) \underbrace{\sum_G \overbrace{\phi_L(L,G)\tau_3(G,S)\tau_4(G,J)}^{\text{(2) product}}}_{\text{(3) marginal: } \tau_5(J,L,S)} = \underbrace{\phi_J(J,L,S)\tau_5(J,L,S)}_{\text{(4) new collection}}$$

VE – Overview of Steps

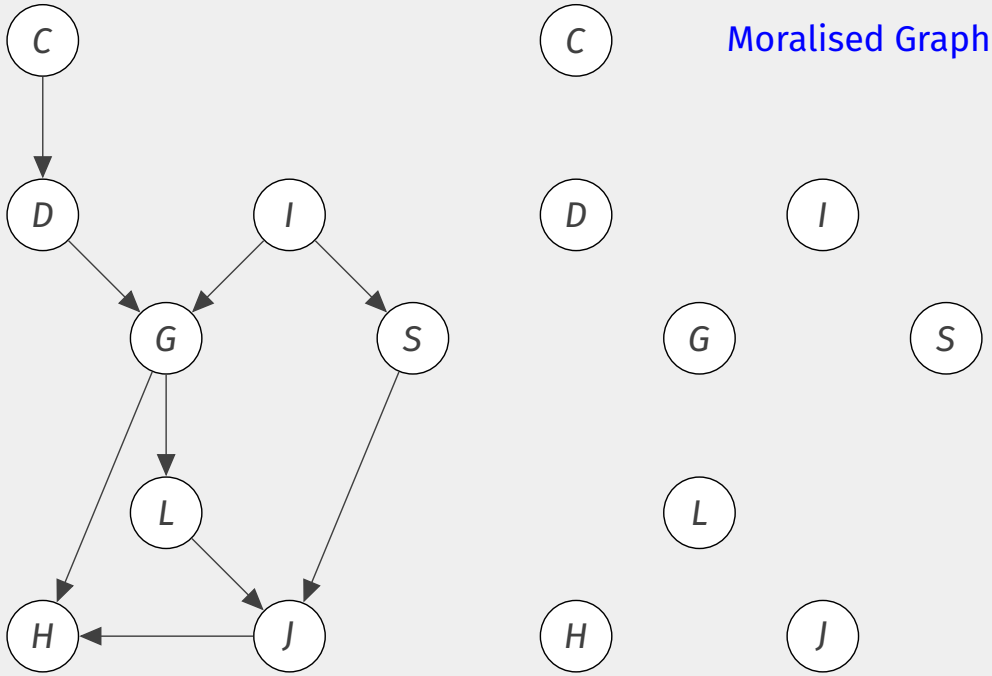
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3. marginalise Y out of the product;
4. add the resulting factor in the collection.

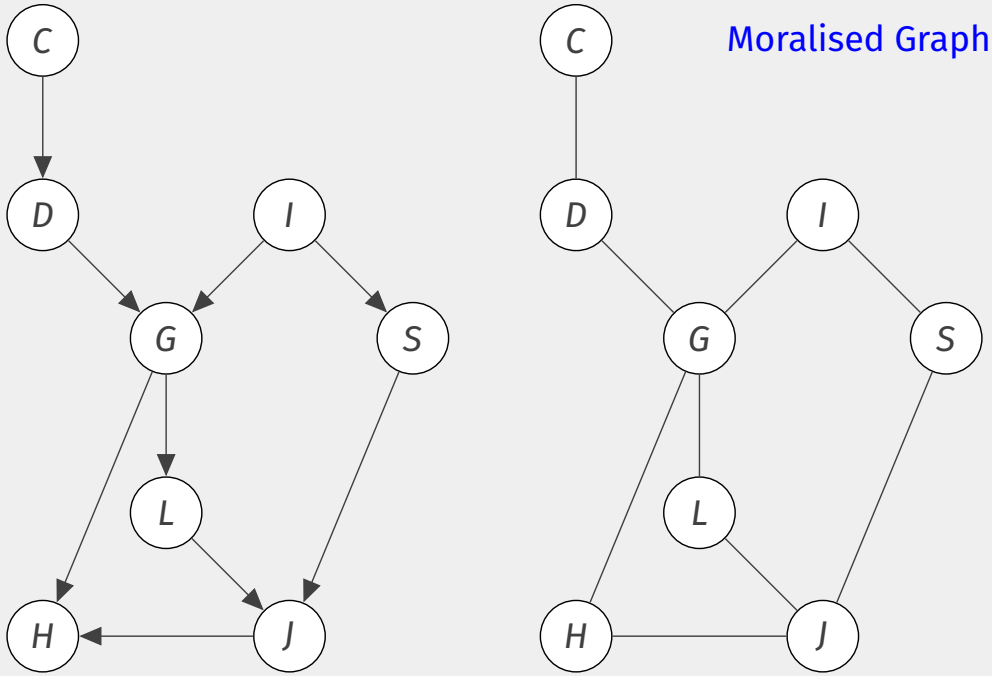
We repeat this until we are done marginalising. The resulting collection induces an MN representation of $\tilde{P}_{\Phi}(\mathbf{Q}, \mathbf{E} = \mathbf{e})$.

Graph View: Moralised Graph *Extended Student*



The **Moralised Graph** [definition 4.16] captures the factor structure of the BN:

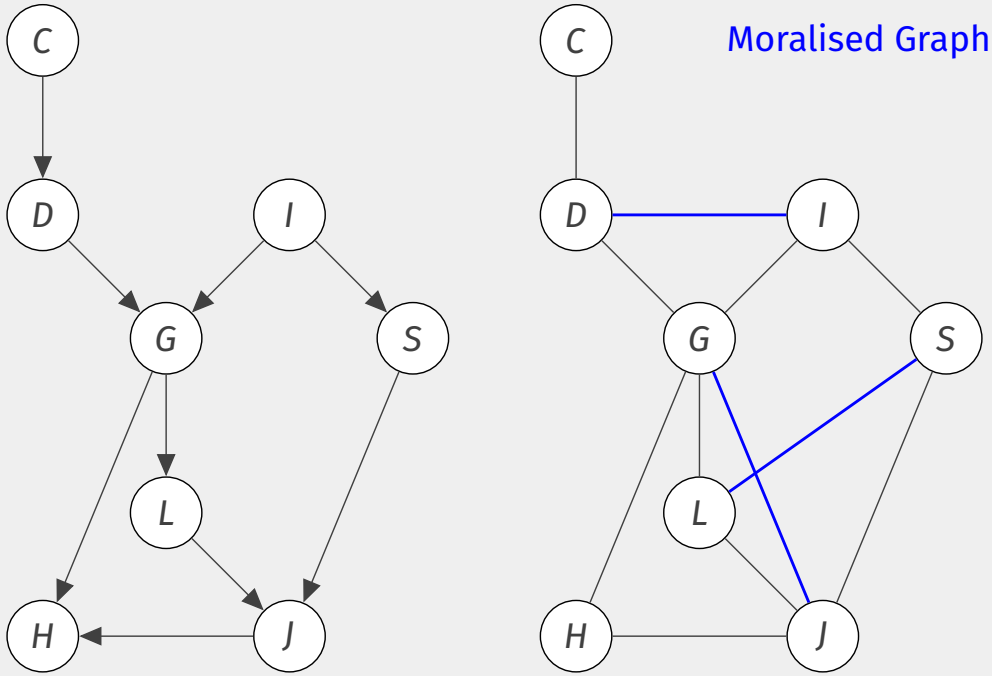
Graph View: Moralised Graph *Extended Student*



The **Moralised Graph** [definition 4.16] captures the factor structure of the BN:

1. make edges undirected

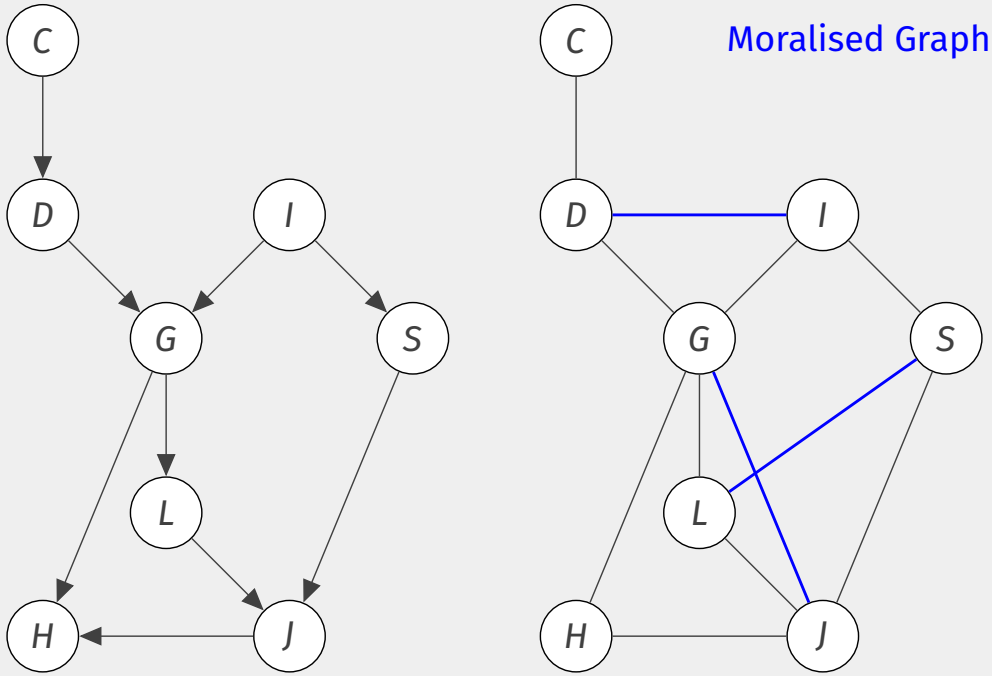
Graph View: Moralised Graph *Extended Student*



The **Moralised Graph** [definition 4.16] captures the factor structure of the BN:

1. make edges undirected
2. '*marry*' the parents of the colliders

Graph View: Moralised Graph *Extended Student*

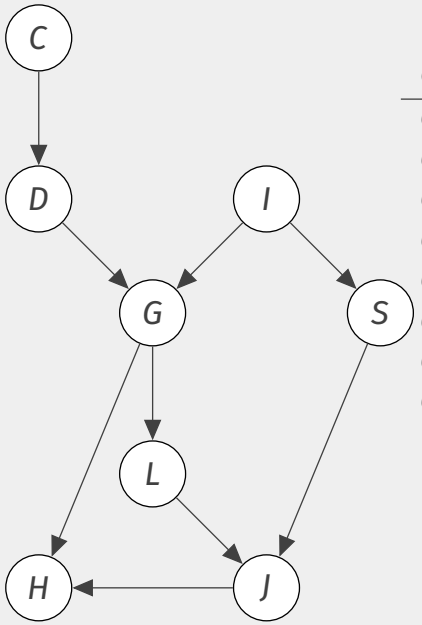


The **Moralised Graph** [definition 4.16] captures the factor structure of the BN:

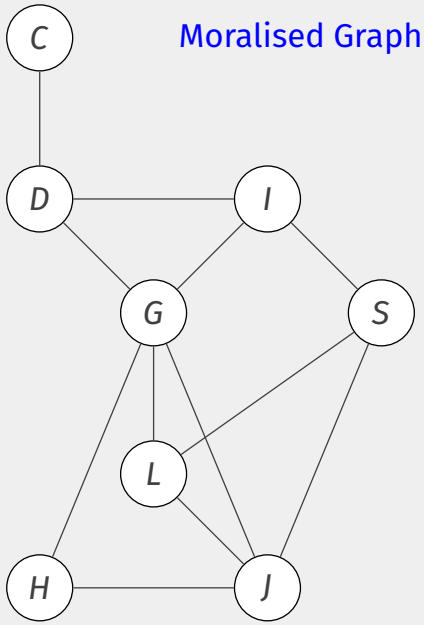
1. make edges undirected
2. *'marry'* the parents of the colliders

If H is the moralised version of the BN structure G, it follows that $I(H) \subseteq I(G)$.
Marginal indep. of parents in a v-structure is missing.
For BNs, VE starts from this graph.

$P(J)$ for *Extended Student* [table 9.1] Graph View

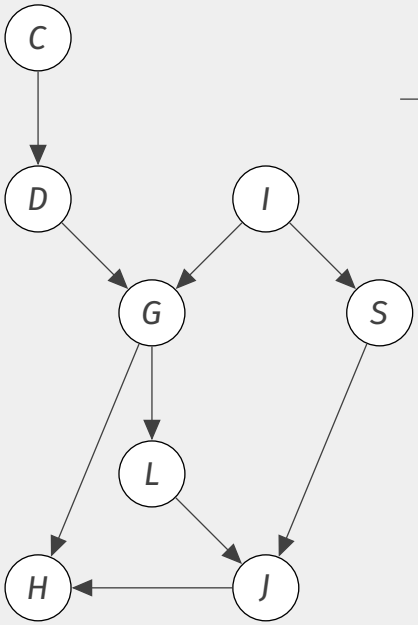


Φ	Scope
ϕ_C	$\underline{C}$
ϕ_D	$\underline{D}, C$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$
ϕ_H	$\underline{H}, G, J$
ϕ_J	$\underline{J}, L, S$

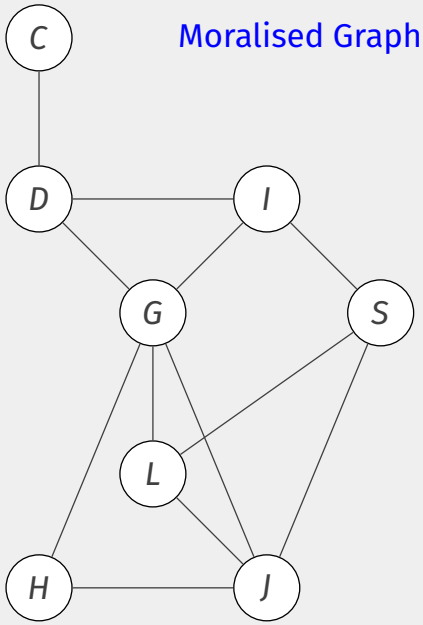


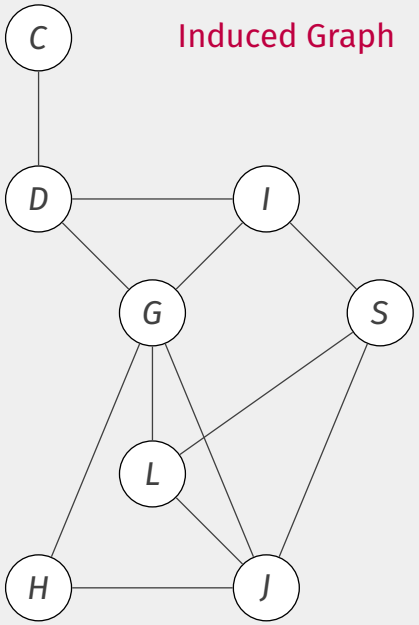
Moralised Graph

$P(J)$ for *Extended Student* [table 9.1] Graph View

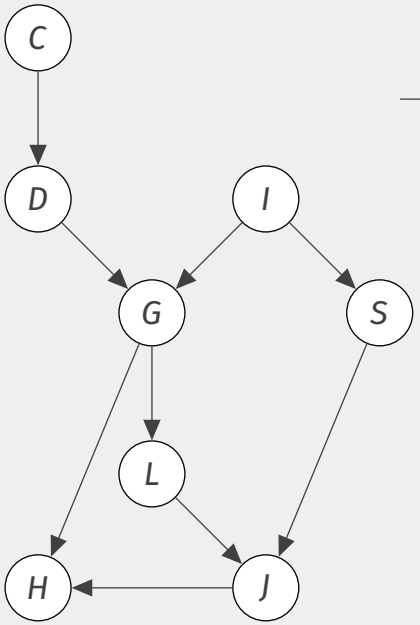


Φ	Scope
ϕ_C	$\underline{C}$
ϕ_D	$\underline{D}, C$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$
ϕ_H	$\underline{H}, G, J$
ϕ_J	$\underline{J}, L, S$

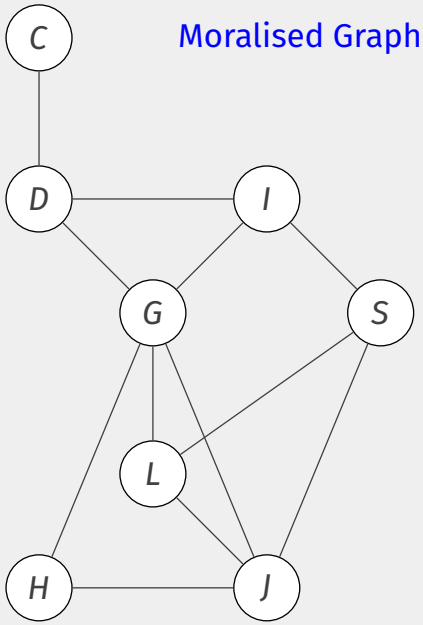


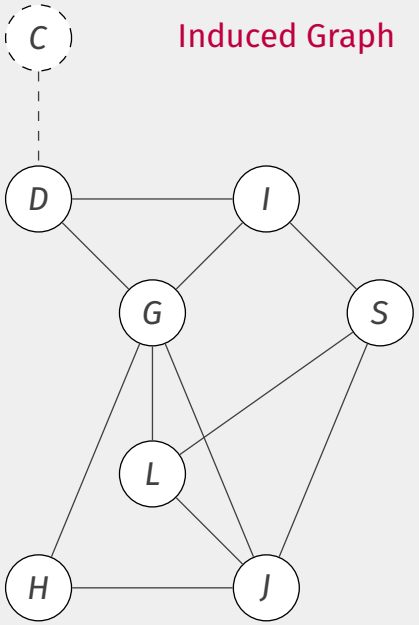


$P(J)$ for *Extended Student* [table 9.1] Graph View



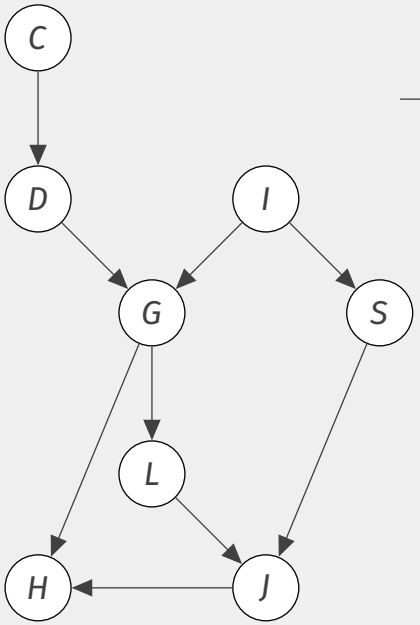
Φ	Scope
ϕ_C	$\underline{C}$
ϕ_D	$\underline{D}, C$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$
ϕ_H	$\underline{H}, G, J$
ϕ_J	$\underline{J}, L, S$



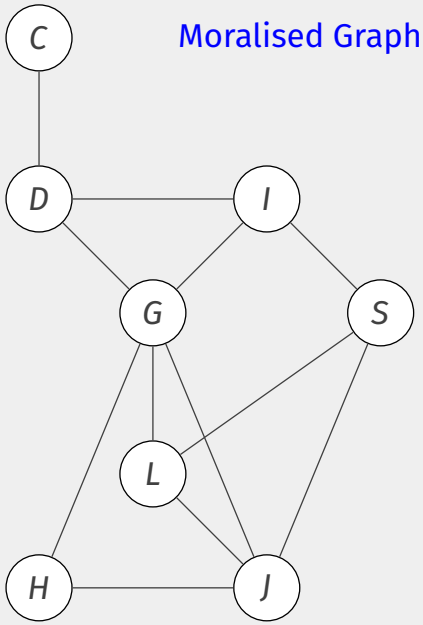


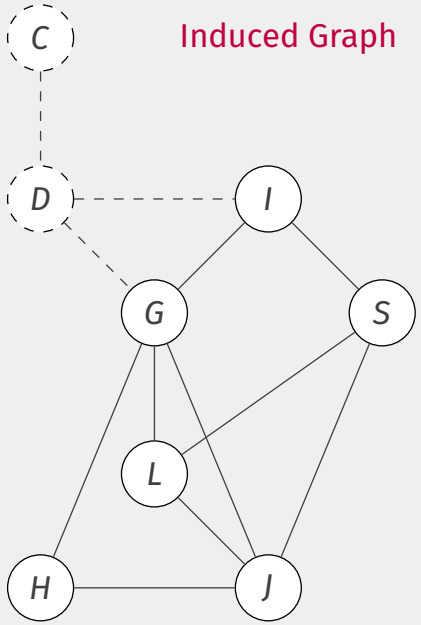
Eliminating	Available factors	Relevant	Intermediate factor	New factor
C	$\phi_C, \phi_D, \phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J$	ϕ_C, ϕ_D	$\rho_1(C, D)$	$\tau_1(D)$

$P(J)$ for *Extended Student* [table 9.1] Graph View



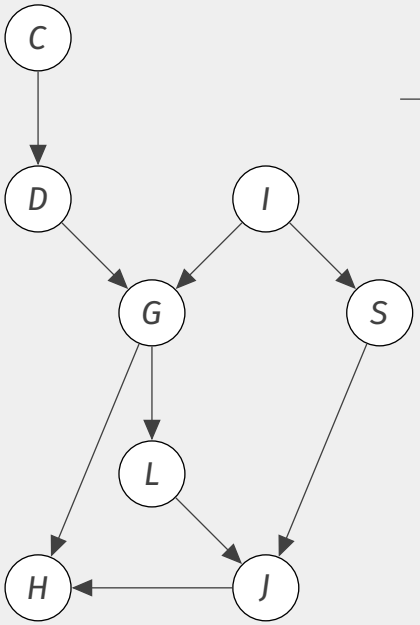
Φ	Scope
ϕ_C	$\underline{C}$
ϕ_D	$\underline{D}, C$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$
ϕ_H	$\underline{H}, G, J$
ϕ_J	$\underline{J}, L, S$



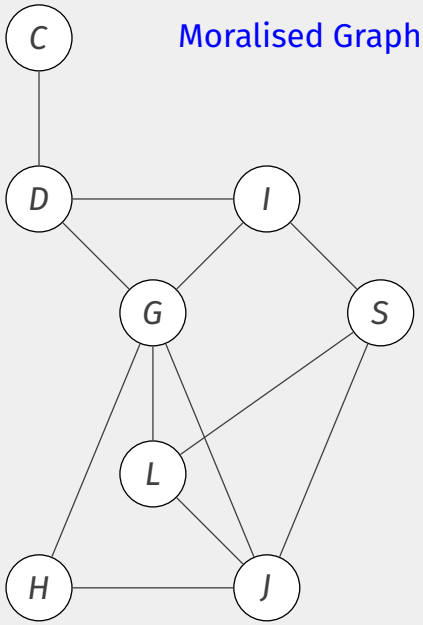


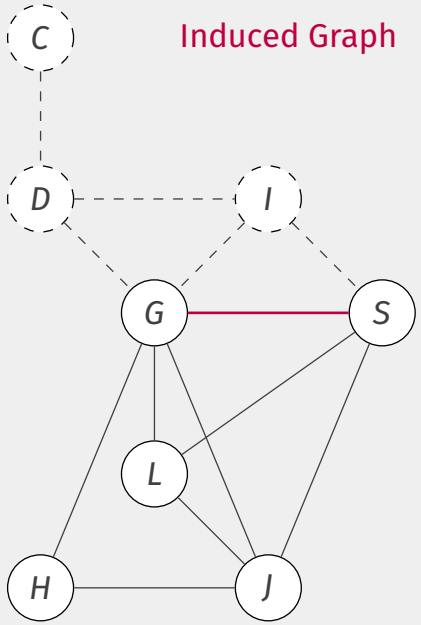
Eliminating	Available factors	Relevant	Intermediate factor	New factor
C	$\phi_C, \phi_D, \phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J$	ϕ_C, ϕ_D	$\rho_1(C, D)$	$\tau_1(D)$
D	$\phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J, \tau_1$	τ_1, ϕ_G	$\rho_2(G, D, I)$	$\tau_2(G, I)$

$P(J)$ for *Extended Student* [table 9.1] Graph View



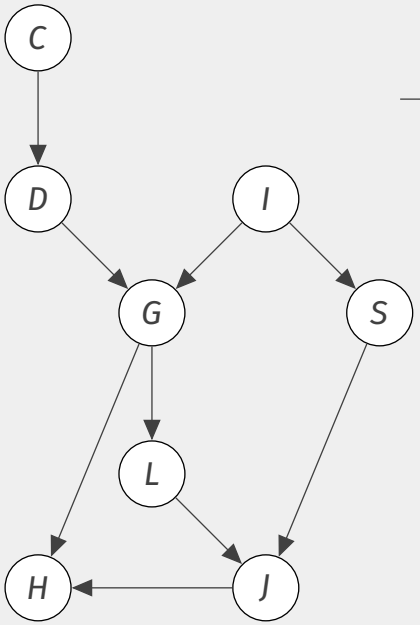
Φ	Scope
ϕ_C	$\underline{C}$
ϕ_D	$\underline{D}, C$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$
ϕ_H	$\underline{H}, G, J$
ϕ_J	$\underline{J}, L, S$



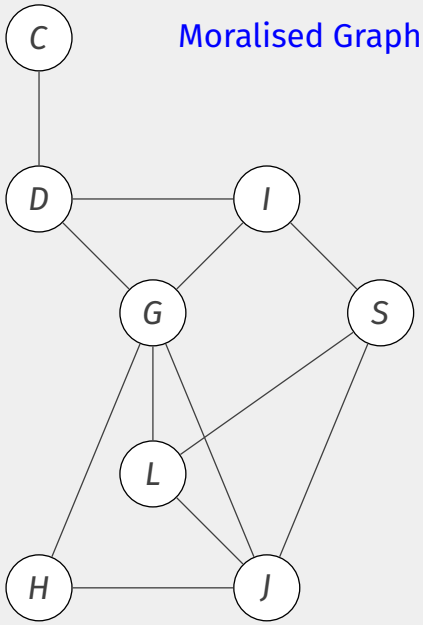


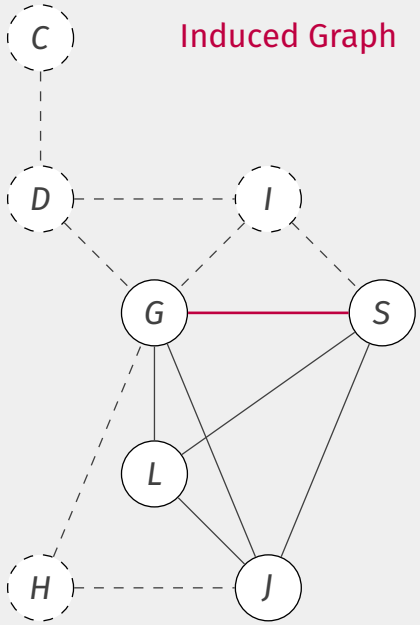
Eliminating	Available factors	Relevant	Intermediate factor	New factor
C	$\phi_C, \phi_D, \phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J$	ϕ_C, ϕ_D	$\rho_1(C, D)$	$\tau_1(D)$
D	$\phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J, \tau_1$	τ_1, ϕ_G	$\rho_2(G, D, I)$	$\tau_2(G, I)$
I	$\phi_I, \phi_S, \phi_L, \phi_H, \phi_J, \tau_2$	ϕ_I, ϕ_S, τ_2	$\rho_3(I, G, S)$	$\tau_3(G, S)$

$P(J)$ for *Extended Student* [table 9.1] Graph View



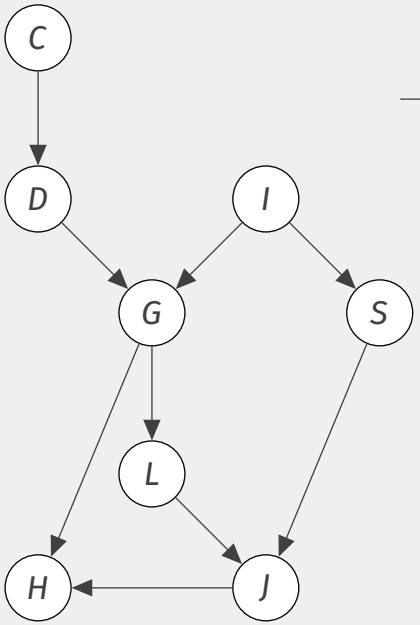
Φ	Scope
ϕ_C	$\underline{C}$
ϕ_D	$\underline{D}, C$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$
ϕ_H	$\underline{H}, G, J$
ϕ_J	$\underline{J}, L, S$



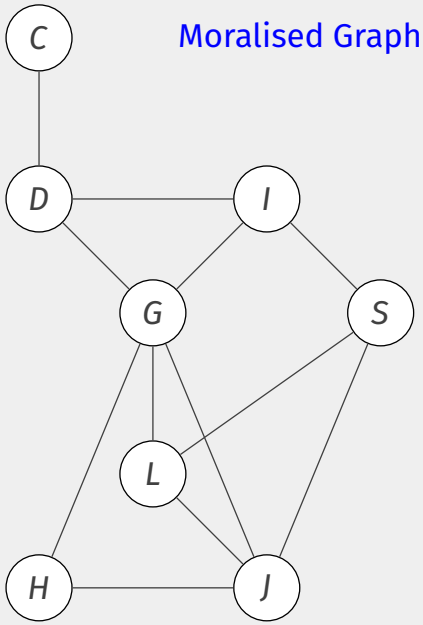


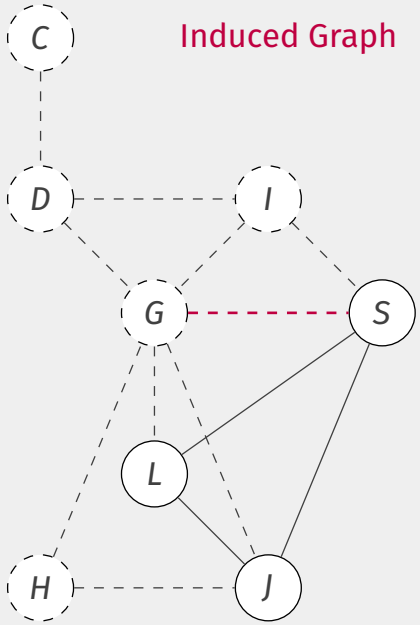
Eliminating	Available factors	Relevant	Intermediate factor	New factor
C	$\phi_C, \phi_D, \phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J$	ϕ_C, ϕ_D	$\rho_1(C, D)$	$\tau_1(D)$
D	$\phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J, \tau_1$	τ_1, ϕ_G	$\rho_2(G, D, I)$	$\tau_2(G, I)$
I	$\phi_I, \phi_S, \phi_L, \phi_H, \phi_J, \tau_2$	ϕ_I, ϕ_S, τ_2	$\rho_3(I, G, S)$	$\tau_3(G, S)$
H	$\phi_L, \phi_H, \phi_J, \tau_3$	ϕ_H	$\phi_H(H, G, J)$	$\tau_4(G, J)$

$P(J)$ for *Extended Student* [table 9.1] Graph View



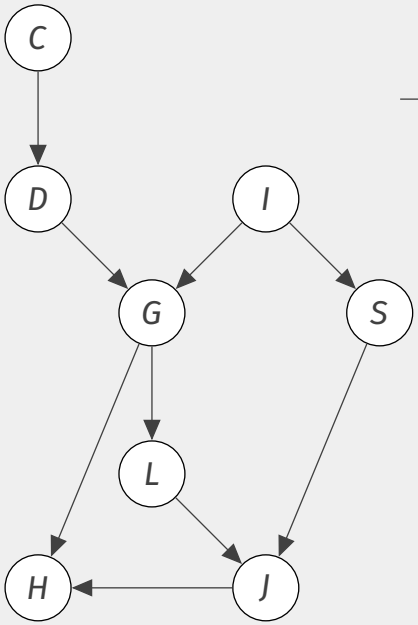
Φ	Scope
ϕ_C	$\underline{C}$
ϕ_D	$\underline{D}, C$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$
ϕ_H	$\underline{H}, G, J$
ϕ_J	$\underline{J}, L, S$



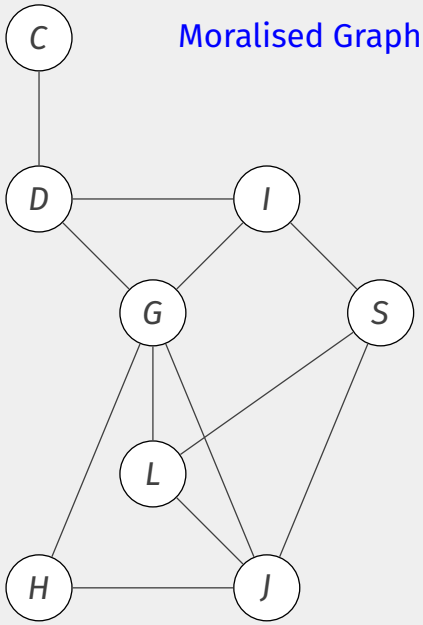


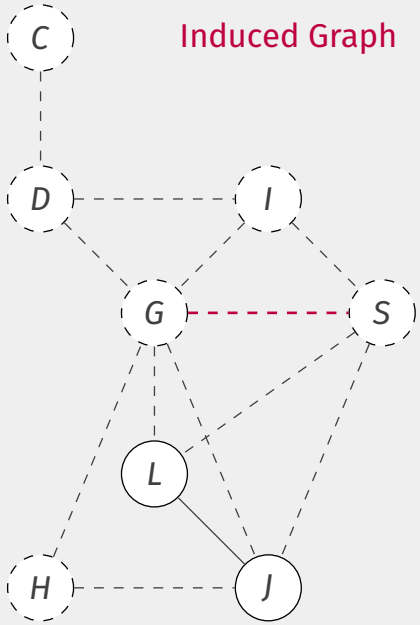
Eliminating	Available factors	Relevant	Intermediate factor	New factor
C	$\phi_C, \phi_D, \phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J$	ϕ_C, ϕ_D	$\rho_1(C, D)$	$\tau_1(D)$
D	$\phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J, \tau_1$	τ_1, ϕ_G	$\rho_2(G, D, I)$	$\tau_2(G, I)$
I	$\phi_I, \phi_S, \phi_L, \phi_H, \phi_J, \tau_2$	ϕ_I, ϕ_S, τ_2	$\rho_3(I, G, S)$	$\tau_3(G, S)$
H	$\phi_L, \phi_H, \phi_J, \tau_3$	ϕ_H	$\phi_H(H, G, J)$	$\tau_4(G, J)$
G	$\phi_L, \phi_J, \tau_3, \tau_4$	ϕ_L, τ_3, τ_4	$\rho_5(G, J, L, S)$	$\tau_5(J, L, S)$

$P(J)$ for *Extended Student* [table 9.1] Graph View



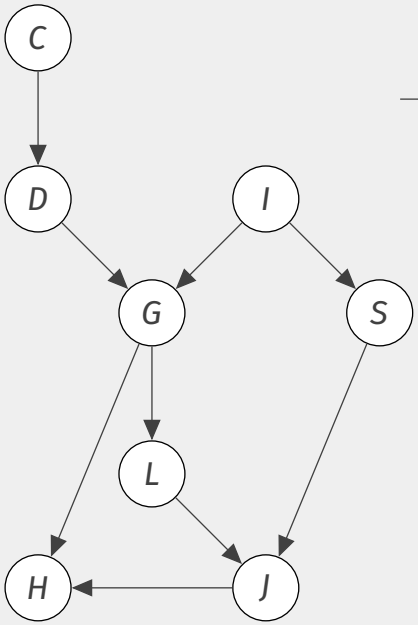
Φ	Scope
ϕ_C	$\underline{C}$
ϕ_D	$\underline{D}, C$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$
ϕ_H	$\underline{H}, G, J$
ϕ_J	$\underline{J}, L, S$



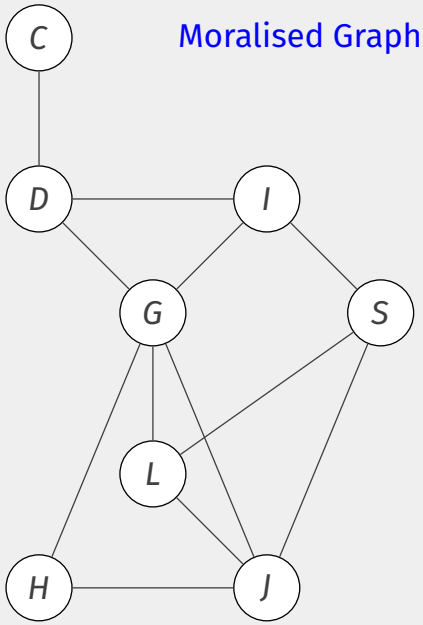


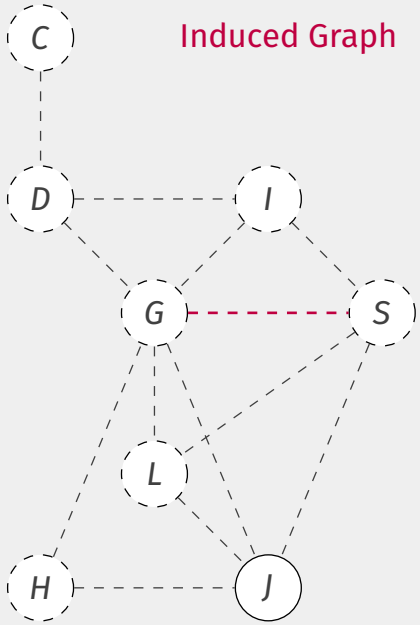
Eliminating	Available factors	Relevant	Intermediate factor	New factor
C	$\phi_C, \phi_D, \phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J$	ϕ_C, ϕ_D	$\rho_1(C, D)$	$\tau_1(D)$
D	$\phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J, \tau_1$	τ_1, ϕ_G	$\rho_2(G, D, I)$	$\tau_2(G, I)$
I	$\phi_I, \phi_S, \phi_L, \phi_H, \phi_J, \tau_2$	ϕ_I, ϕ_S, τ_2	$\rho_3(I, G, S)$	$\tau_3(G, S)$
H	$\phi_L, \phi_H, \phi_J, \tau_3$	ϕ_H	$\phi_H(H, G, J)$	$\tau_4(G, J)$
G	$\phi_L, \phi_J, \tau_3, \tau_4$	ϕ_L, τ_3, τ_4	$\rho_5(G, J, L, S)$	$\tau_5(J, L, S)$
S	ϕ_J, τ_5	ϕ_J, τ_5	$\rho_6(J, L, S)$	$\tau_6(J, L)$

$P(J)$ for *Extended Student* [table 9.1] Graph View

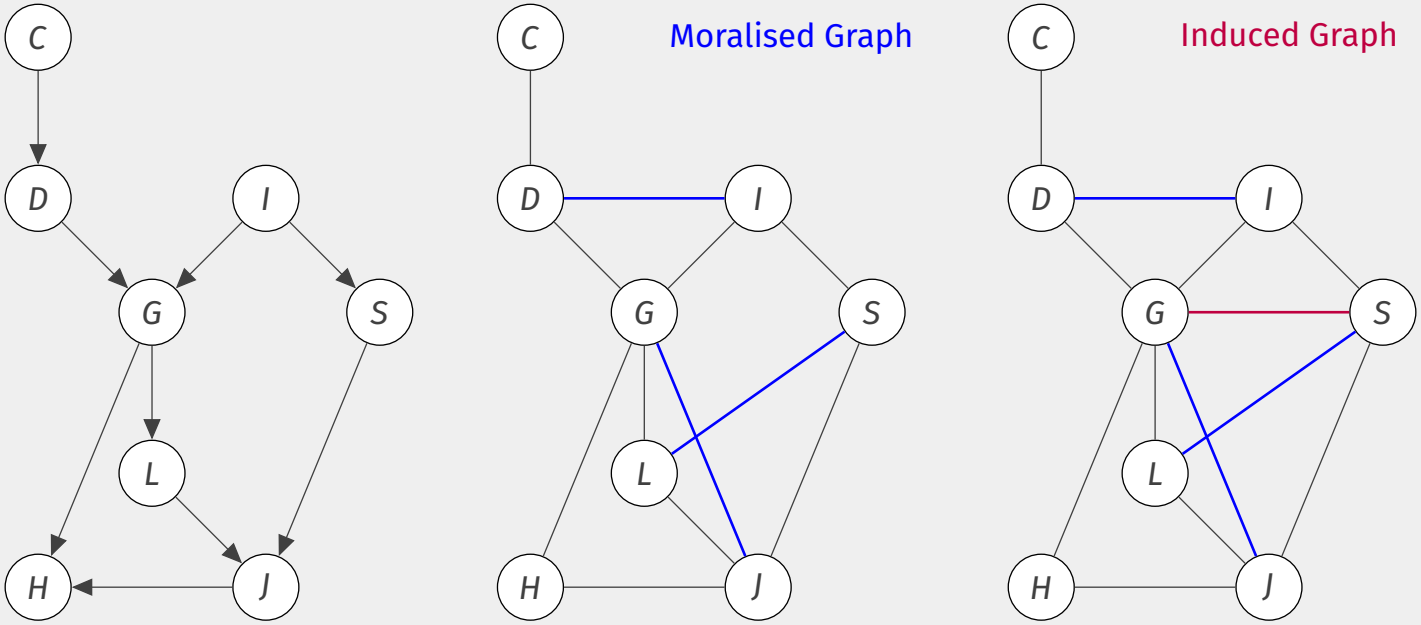


Φ	Scope
ϕ_C	$\underline{C}$
ϕ_D	$\underline{D}, C$
ϕ_I	$\underline{I}$
ϕ_G	$\underline{G}, D, I$
ϕ_S	$\underline{S}, I$
ϕ_L	$\underline{L}, G$
ϕ_H	$\underline{H}, G, J$
ϕ_J	$\underline{J}, L, S$





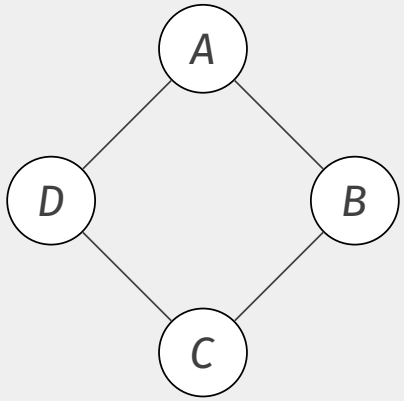
Eliminating	Available factors	Relevant	Intermediate factor	New factor
C	$\phi_C, \phi_D, \phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J$	ϕ_C, ϕ_D	$\rho_1(C, D)$	$\tau_1(D)$
D	$\phi_I, \phi_S, \phi_G, \phi_L, \phi_H, \phi_J, \tau_1$	τ_1, ϕ_G	$\rho_2(G, D, I)$	$\tau_2(G, I)$
I	$\phi_I, \phi_S, \phi_L, \phi_H, \phi_J, \tau_2$	ϕ_I, ϕ_S, τ_2	$\rho_3(I, G, S)$	$\tau_3(G, S)$
H	$\phi_L, \phi_H, \phi_J, \tau_3$	ϕ_H	$\phi_H(H, G, J)$	$\tau_4(G, J)$
G	$\phi_L, \phi_J, \tau_3, \tau_4$	ϕ_L, τ_3, τ_4	$\rho_5(G, J, L, S)$	$\tau_5(J, L, S)$
S	ϕ_J, τ_5	ϕ_J, τ_5	$\rho_6(J, L, S)$	$\tau_6(J, L)$
L	τ_6	τ_6	$\tau_6(J, L)$	$\tau_7(J)$



Elimination of Y as graph transformation [section 9.4.2.2]:

1. connect all neighbours of Y to one another—this captures the marginal factor that we add to the collection; depending on elimination order, this may create new edges called 'fill edges';
2. remove Y and all of its incident edges from the graph;

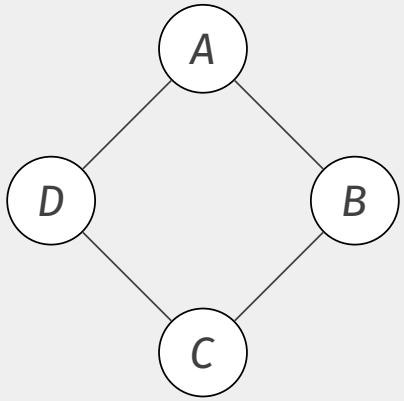
The **Induced Graph** [definition 9.5] is the union of all intermediate graphs.
Exercise: elimination order G, I, S, L, H, C, D [solution in table 9.2].



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

Joint unnormalised distribution:

$$\tilde{P}_\Phi(A, B, C, D) = \phi_1(A, B)\phi_2(B, C)\phi_3(C, D)\phi_4(D, A)$$

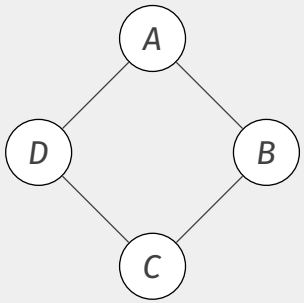


Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

Joint unnormalised distribution:

$$\tilde{P}_\Phi(A, B, C, D) = \phi_1(A, B)\phi_2(B, C)\phi_3(C, D)\phi_4(D, A)$$

Example: let's express $P_\Phi(D)$ by eliminating A, B, C in this order



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

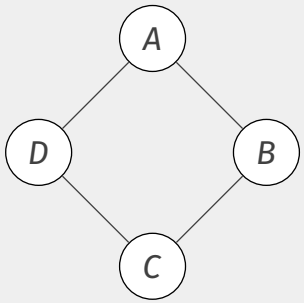
Joint unnormalised distribution:

$$\tilde{P}_\Phi(A, B, C, D) = \phi_1(A, B)\phi_2(B, C)\phi_3(C, D)\phi_4(D, A)$$

Goal: $P_\Phi(D)$.

Eliminate: $\underline{A}, B, C$.

$$\sum_A \phi_1(A, B)\phi_2(B, C)\phi_3(C, D)\phi_4(D, A)$$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

Joint unnormalised distribution:

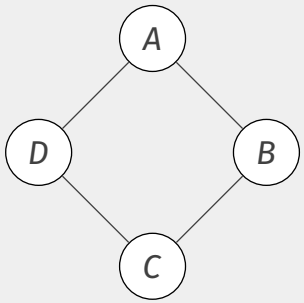
$$\tilde{P}_\Phi(A,B,C,D) = \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

Goal: $P_\Phi(D)$.

Eliminate: $\underline{A}, B, C$.

$$\sum_A \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

$$= \phi_2(B,C)\phi_3(C,D) \underbrace{\sum_A \overbrace{\phi_1(A,B)\phi_4(D,A)}^{\rho_1(A,B,D)}}_{\tau_1(B,D)}$$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

Joint unnormalised distribution:

$$\tilde{P}_\Phi(A,B,C,D) = \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

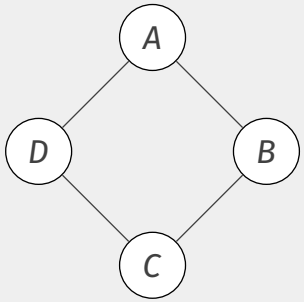
Goal: $P_\Phi(D)$.

Eliminate: A, B, C.

$$\sum_A \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

$$= \phi_2(B,C)\phi_3(C,D) \underbrace{\sum_A \overbrace{\phi_1(A,B)\phi_4(D,A)}^{\rho_1(A,B,D)}}_{\tau_1(B,D)}$$

$$= \phi_2(B,C)\phi_3(C,D)\tau_1(B,D)$$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

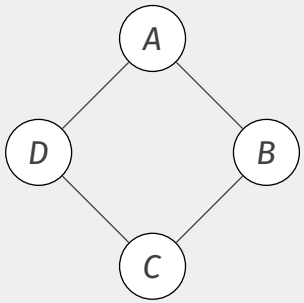
Eliminate: ~~A~~, $\underline{B}$, C.

Joint unnormalised distribution:

$$\tilde{P}_\Phi(A,B,C,D) = \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

Goal: $P_\Phi(D)$.

$$\sum_B \phi_2(B,C)\phi_3(C,D)\tau_1(B,D)$$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

Eliminate: ~~A~~, $\underline{B}$, C.

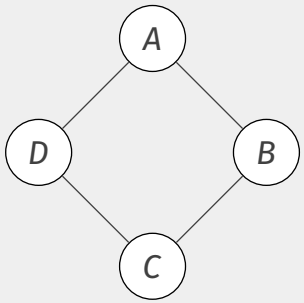
Joint unnormalised distribution:

$$\tilde{P}_\Phi(A,B,C,D) = \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

Goal: $P_\Phi(D)$.

$$\sum_B \phi_2(B,C)\phi_3(C,D)\tau_1(B,D)$$

$$= \phi_3(C,D) \underbrace{\sum_B \overbrace{\phi_2(B,C)\tau_1(B,D)}^{\rho_2(B,C,D)}}_{\tau_2(C,D)}$$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

Joint unnormalised distribution:

$$\tilde{P}_\Phi(A,B,C,D) = \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

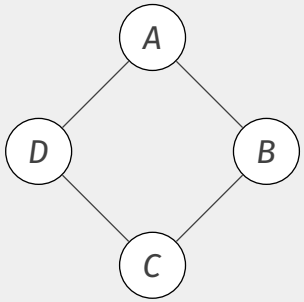
Goal: $P_\Phi(D)$.

Eliminate: ~~A~~, B, C.

$$\sum_B \phi_2(B,C)\phi_3(C,D)\tau_1(B,D)$$

$$= \phi_3(C,D) \underbrace{\sum_B \overbrace{\phi_2(B,C)\tau_1(B,D)}^{\rho_2(B,C,D)}}_{\tau_2(C,D)}$$

$$= \phi_3(C,D)\tau_2(C,D)$$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

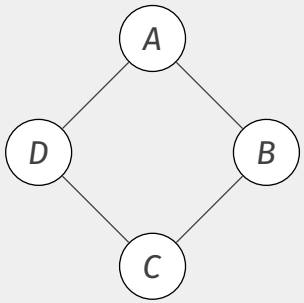
Eliminate: ~~A~~, ~~B~~, C.

Joint unnormalised distribution:

$$\tilde{P}_\Phi(A, B, C, D) = \phi_1(A, B)\phi_2(B, C)\phi_3(C, D)\phi_4(D, A)$$

Goal: $P_\Phi(D)$.

$$\sum_C \phi_3(C, D)\tau_2(C, D)$$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

Eliminate: ~~A~~, ~~B~~, C.

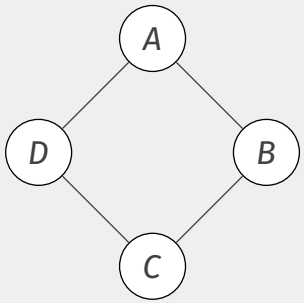
Joint unnormalised distribution:

$$\tilde{P}_\Phi(A,B,C,D) = \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

Goal: $P_\Phi(D)$.

$$\sum_C \phi_3(C,D)\tau_2(C,D)$$

$$= \sum_C \overbrace{\phi_3(C,D)\tau_2(C,D)}^{\rho_3(C,D)}$$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
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Eliminate: ~~A~~, ~~B~~, C.

Joint unnormalised distribution:

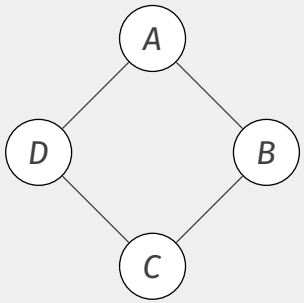
$$\tilde{P}_\Phi(A,B,C,D) = \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

Goal: $P_\Phi(D)$.

$$\sum_C \phi_3(C,D)\tau_2(C,D)$$

$$= \sum_C \overbrace{\phi_3(C,D)\tau_2(C,D)}^{\rho_3(C,D)}$$

$$= \tau_3(D)$$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

Joint unnormalised distribution:

$$\tilde{P}_\Phi(A,B,C,D) = \phi_1(A,B)\phi_2(B,C)\phi_3(C,D)\phi_4(D,A)$$

Goal: $P_\Phi(D)$.

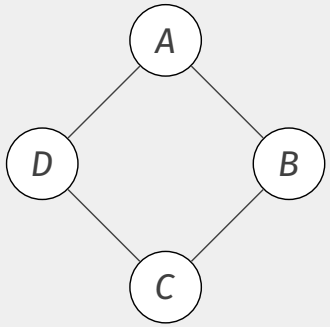
Eliminate: ~~A~~, ~~B~~, C.

$$\sum_C \phi_3(C,D)\tau_2(C,D)$$

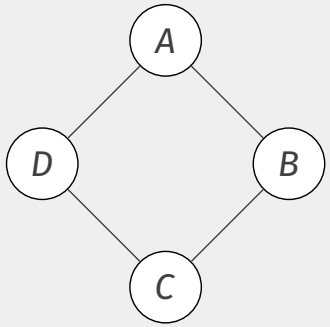
$$= \sum_C \overbrace{\phi_3(C,D)\tau_2(C,D)}^{\rho_3(C,D)}$$

$$= \tau_3(D)$$

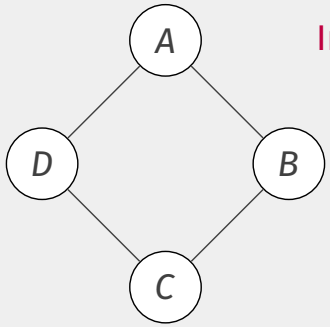
$\tau_3(D)$ is $\tilde{P}_\Phi(D)$.
To obtain $P_\Phi(D)$ we normalise $\tilde{P}(D)$ using the normaliser $\sum_{d\in\mathrm{Val}(D)} \tau_3(D=d)$.



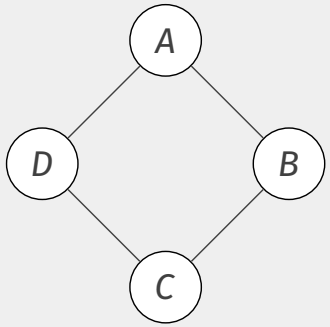
Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A



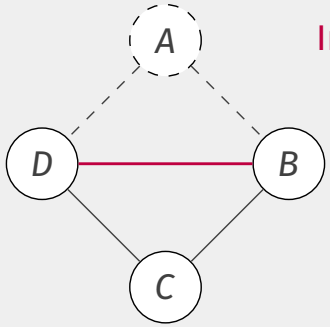
Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A



Induced Graph

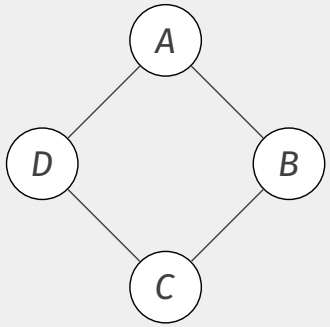


Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

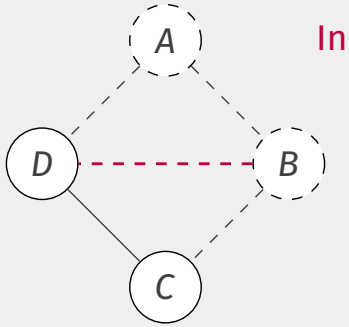


Induced Graph

Eliminating	Available factors	Relevant	Intermediate factor	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D)$

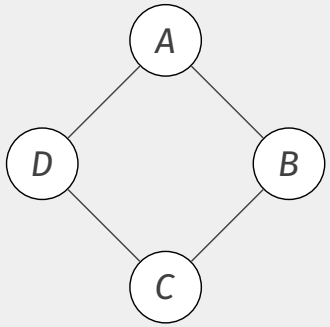


Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

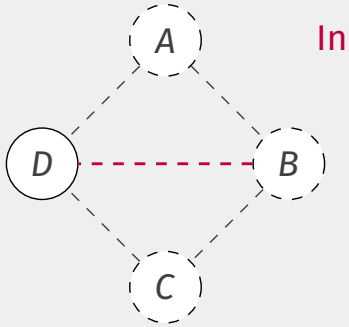


Induced Graph

Eliminating	Available factors	Relevant	Intermediate factor	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D)$
B	ϕ_2, ϕ_3, τ_1	ϕ_2, τ_1	$\rho_2(B, C, D)$	$\tau_2(C, D)$

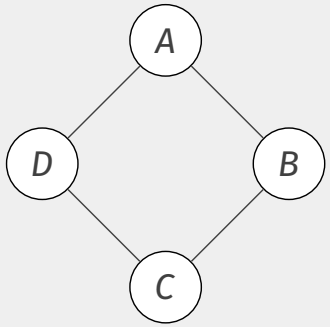


Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
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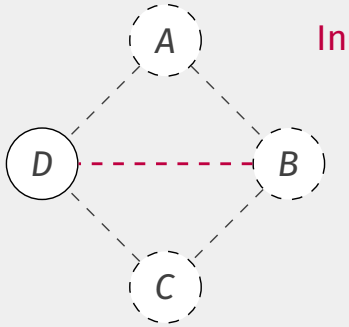


Induced Graph

Eliminating	Available factors	Relevant	Intermediate factor	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D)$
B	ϕ_2, ϕ_3, τ_1	ϕ_2, τ_1	$\rho_2(B, C, D)$	$\tau_2(C, D)$
C	ϕ_3, τ_2	ϕ_3, τ_2	$\rho_3(C, D)$	$\tau_3(D)$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A



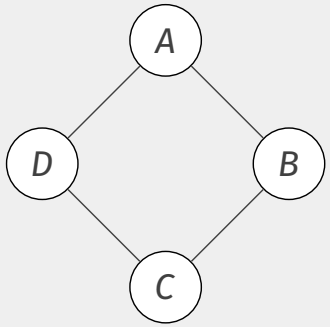
Induced Graph

Eliminating	Available factors	Relevant	Intermediate factor	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D)$
B	ϕ_2, ϕ_3, τ_1	ϕ_2, τ_1	$\rho_2(B, C, D)$	$\tau_2(C, D)$
C	ϕ_3, τ_2	ϕ_3, τ_2	$\rho_3(C, D)$	$\tau_3(D)$

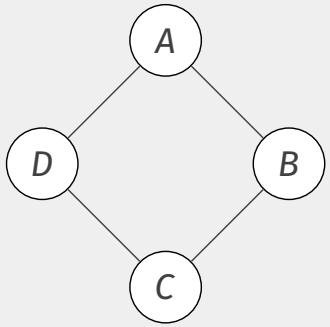
There's no need for 'moralisation' because MNs do not have v-structures.
Elimination of Y as graph transformation [section 9.4.2.2]:

1. connect all neighbours of Y to one another;
2. remove Y and all of its incident edges from the graph;

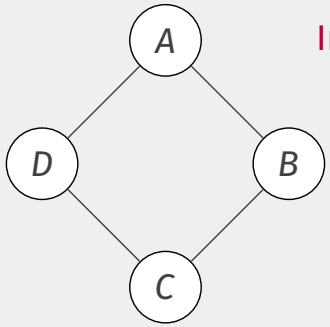
The **Induced Graph** [definition 9.5] is the union of all intermediate graphs.
Exercise: VE in order B, C, A.



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

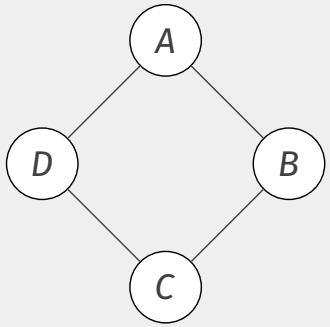


Φ	Scope
ϕ_1	A, B
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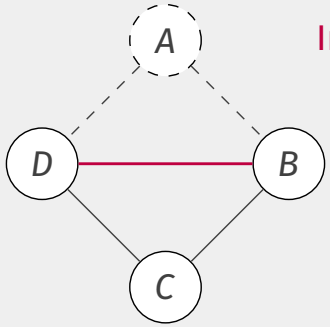


Induced Graph

$P_\Phi(B, C, D)$ for *Misconception* — Graph View

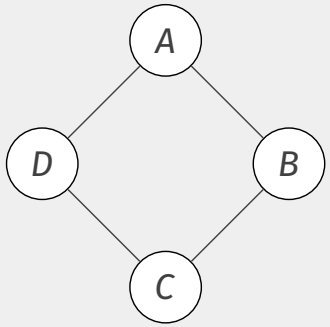


Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

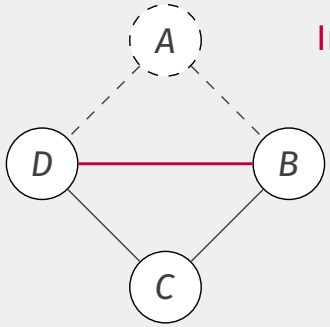


Induced Graph

Eliminating	Available factors	Relevant	Intermediate factor	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D)$



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A



Induced Graph

Eliminating	Available factors	Relevant	Intermediate factor	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D)$

The final graph is an MN with

- 3 nodes B, C, D,
- 3 edges (B–C and C–D from the original MN and the fill edge B–D),
- and 3 factors (ϕ_2, ϕ_4 from the original MN and the new factor τ_1).

VE Algorithm [Sec 9.3; Algorithms 9.1 and 9.2]

Start with a network with nodes $\mathbf{X}$ and factors Φ
(for a BN, use the moralised graph)

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If there's evidence $\mathbf{E} = \mathbf{e}$, reduce all factors in Φ. [Section 9.3.2]

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For a query $\mathbf{Q}$, disjoint with $\mathbf{E}$, we need to marginalise $\mathbf{M} = \mathbf{X} \setminus (\mathbf{Q} \cup \mathbf{E})$.

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In a given order, for each rv $Y \in \mathbf{M}$ that we want to marginalise, we

1. gather all relevant factors—those for which Y is in the scope—removing them from Φ;
2. take the product of the relevant factors;
3. marginalise Y out of the product;
4. add the resulting factor in the collection.

The remaining factors induce an MN representation of $\tilde{P}_\Phi(\mathbf{Q}, \mathbf{E} = \mathbf{e})$.

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Start with a network with nodes $\mathbf{X}$ and factors Φ
(for a BN, use the moralised graph)

If there's evidence $\mathbf{E} = \mathbf{e}$, reduce all factors in Φ. [Section 9.3.2]

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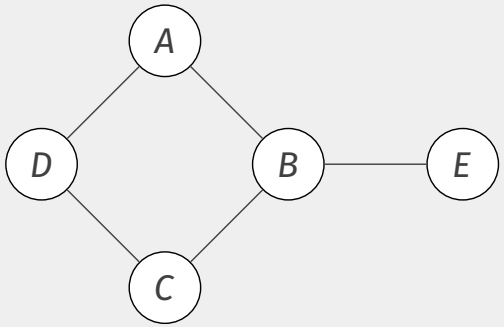
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1. gather all relevant factors—those for which Y is in the scope—removing them from Φ;
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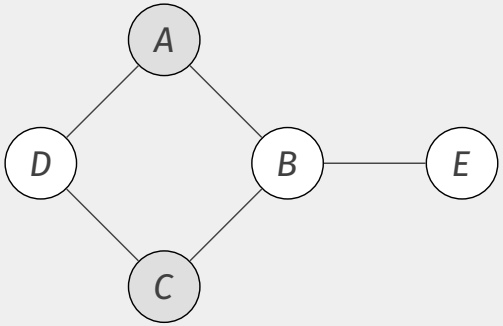
For a tabular view: multiply remaining factors and normalise.

VE with Evidence: $P_\Phi(D|A = a^1, C = c^1)$ Graph View



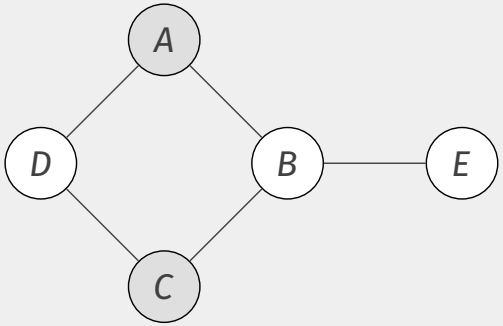
Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A
ϕ_5	B, E

VE with Evidence: $P_\Phi(D|A = a^1, C = c^1)$ Graph View

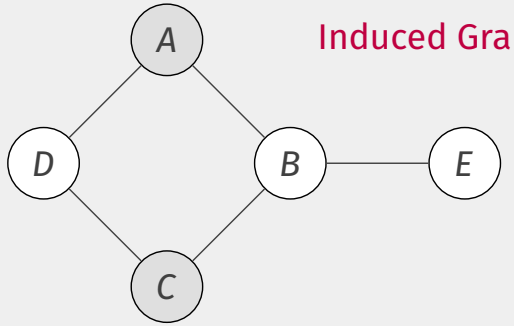


Φ	Scope
$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
$\phi_3[C = c^1]$	D
$\phi_4[A = a^1]$	D
ϕ_5	B, E

VE with Evidence: $P_\Phi(D|A = a^1, C = c^1)$ Graph View

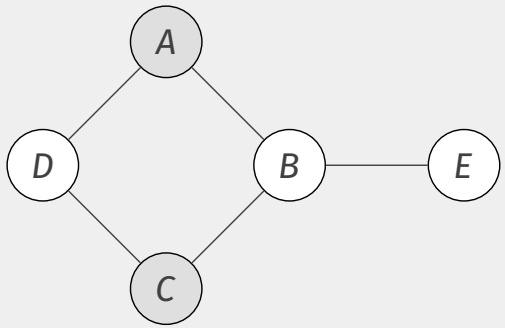


Φ	Scope
$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
$\phi_3[C = c^1]$	D
$\phi_4[A = a^1]$	D
ϕ_5	B, E

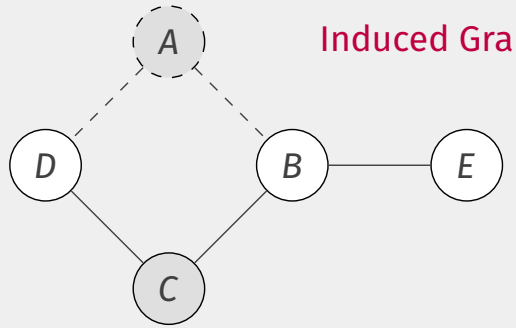


Induced Graph

VE with Evidence: $P_\Phi(D|A = a^1, C = c^1)$ Graph View



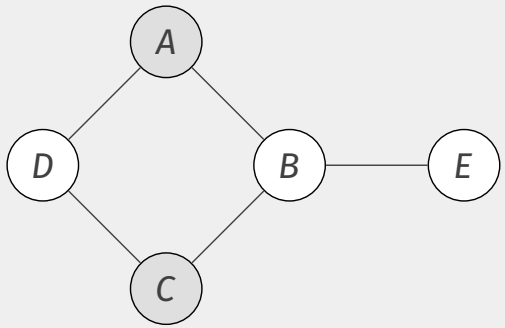
Φ	Scope
$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
$\phi_3[C = c^1]$	D
$\phi_4[A = a^1]$	D
ϕ_5	B, E



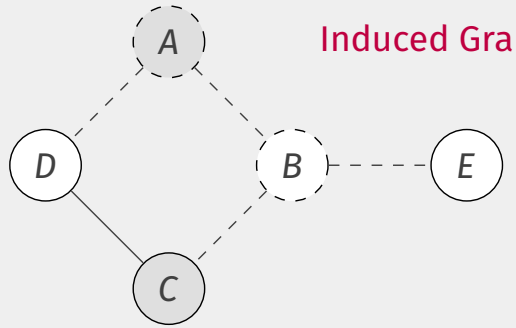
Induced Graph

Elim.	Available factors	Relevant	Intermediate	New factor
A	$\phi_1[a^1], \phi_2[c^1], \phi_3[c^1], \phi_4[a^1], \phi_5$	$\emptyset$	-	-

VE with Evidence: $P_\Phi(D|A = a^1, C = c^1)$ Graph View



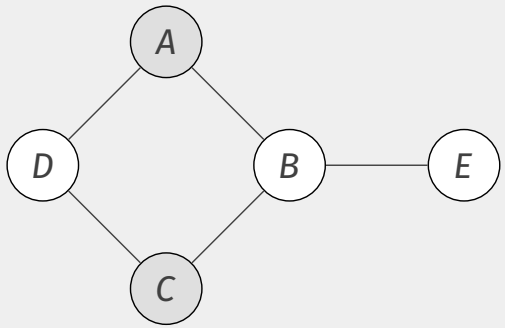
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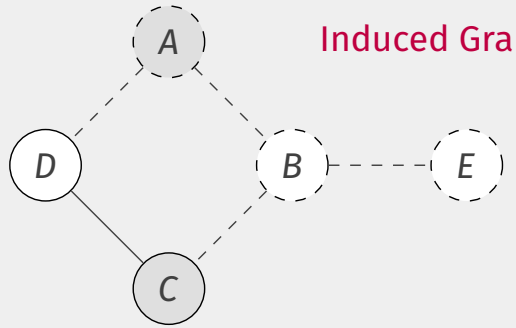
Induced Graph

Elim.	Available factors	Relevant	Intermediate	New factor
A	$\phi_1[a^1], \phi_2[c^1], \phi_3[c^1], \phi_4[a^1], \phi_5$	$\emptyset$	-	-
B	$\phi_1[a^1], \phi_2[c^1], \phi_3[c^1], \phi_4[a^1], \phi_5$	$\phi_1[a^1], \phi_2[c^1], \phi_5$	$\rho_2(B, E)$	$\tau_2(E)$

VE with Evidence: $P_\Phi(D|A = a^1, C = c^1)$ Graph View



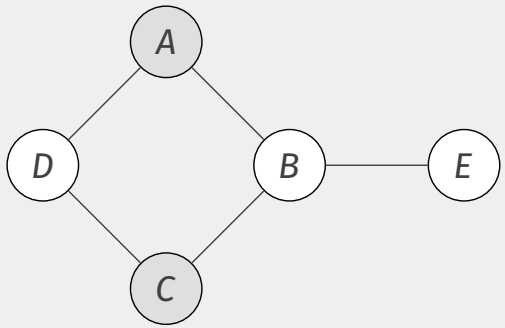
Φ	Scope
$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
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$\phi_4[A = a^1]$	D
ϕ_5	B, E



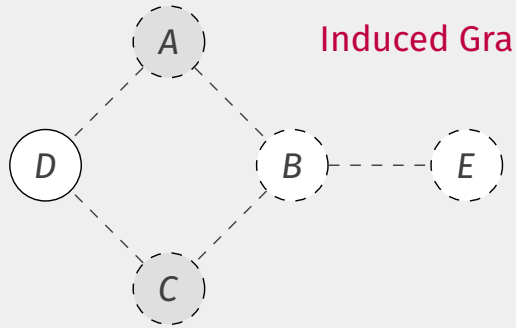
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A	$\phi_1[a^1], \phi_2[c^1], \phi_3[c^1], \phi_4[a^1], \phi_5$	$\emptyset$	-	-
B	$\phi_1[a^1], \phi_2[c^1], \phi_3[c^1], \phi_4[a^1], \phi_5$	$\phi_1[a^1], \phi_2[c^1], \phi_5$	$\rho_2(B, E)$	$\tau_2(E)$
E	$\phi_3[c^1], \phi_4[a^1], \tau_2$	τ_2	$\tau_2(E)$	γ_3

VE with Evidence: $P_\Phi(D|A = a^1, C = c^1)$ Graph View



Φ	Scope
$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
$\phi_3[C = c^1]$	D
$\phi_4[A = a^1]$	D
ϕ_5	B, E

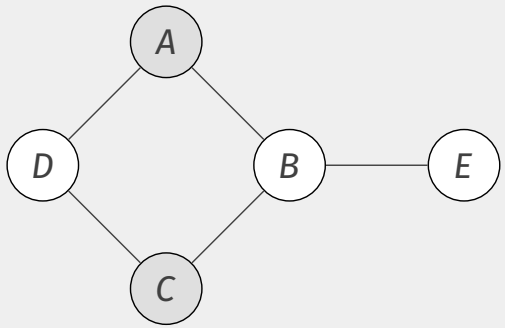


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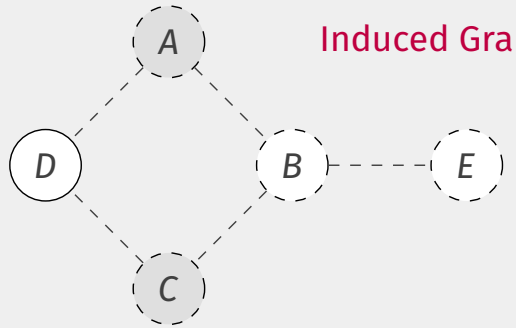
Elim.	Available factors	Relevant	Intermediate	New factor
A	$\phi_1[a^1], \phi_2[c^1], \phi_3[c^1], \phi_4[a^1], \phi_5$	$\emptyset$	-	-
B	$\phi_1[a^1], \phi_2[c^1], \phi_3[c^1], \phi_4[a^1], \phi_5$	$\phi_1[a^1], \phi_2[c^1], \phi_5$	$\rho_2(B, E)$	$\tau_2(E)$
E	$\phi_3[c^1], \phi_4[a^1], \tau_2$	τ_2	$\tau_2(E)$	γ_3
C	$\phi_3[c^1], \phi_4[a^1]\gamma_3$	$\emptyset$	-	-

$$\tilde{P}_\Phi(D, A = a^1, C = c^1) = \phi_3[c^1](D)\phi_4[a^1](D)\gamma_3$$

VE with Evidence: $P_\Phi(D|A = a^1, C = c^1)$ Graph View



Φ	Scope
$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
$\phi_3[C = c^1]$	D
$\phi_4[A = a^1]$	D
ϕ_5	B, E



Induced Graph

Elim.	Available factors	Relevant	Intermediate	New factor
A	$\phi_1[a^1], \phi_2[c^1], \phi_3[c^1], \phi_4[a^1], \phi_5$	$\emptyset$	-	-
B	$\phi_1[a^1], \phi_2[c^1], \phi_3[c^1], \phi_4[a^1], \phi_5$	$\phi_1[a^1], \phi_2[c^1], \phi_5$	$\rho_2(B, E)$	$\tau_2(E)$
E	$\phi_3[c^1], \phi_4[a^1], \tau_2$	τ_2	$\tau_2(E)$	γ_3
C	$\phi_3[c^1], \phi_4[a^1]\gamma_3$	$\emptyset$	-	-

$$\tilde{P}_\Phi(D, A = a^1, C = c^1) = \phi_3[c^1](D)\phi_4[a^1](D)\gamma_3$$

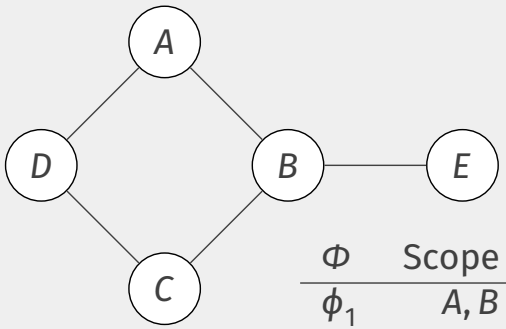
To obtain $P_\Phi(D|A = a^1, C = c^1)$ we divide $\tilde{P}_\Phi(D, A = a^1, C = c^1)$ by the normaliser $\sum_{d \in \text{Val}(D)} \phi_3[c^1](D = d)\phi_4[a^1](D = d)\gamma_3$.

Graphical Separation

We can use graphical separation to spare some unnecessary computation in VE.

Key: if an rv Y to be eliminated is **separate** from $\mathbf{Q}$ given $\mathbf{E}$, then we can eliminate Y without multiplying the factors that involve Y.

Example: $P_\Phi(D|A = a^1, C = c^1)$ in *Misconception MN*



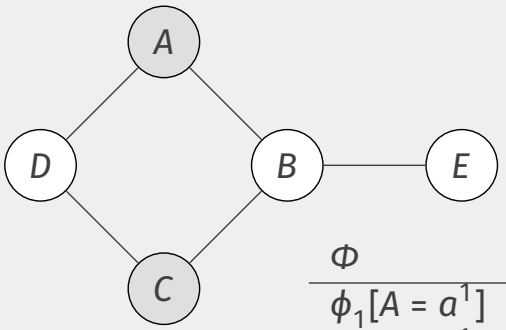
Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A
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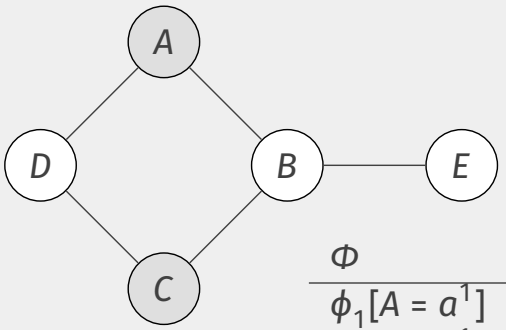
Φ	Scope
$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
$\phi_3[C = c^1]$	D
$\phi_4[A = a^1]$	D
ϕ_5	B, E

Graphical Separation

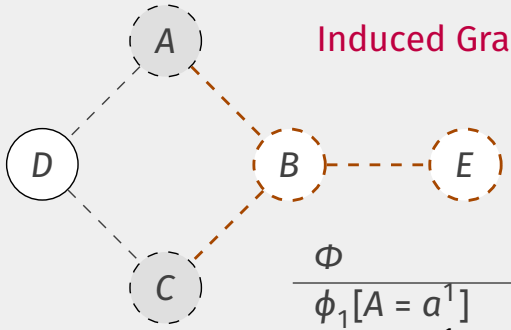
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$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
$\phi_3[C = c^1]$	D
$\phi_4[A = a^1]$	D
ϕ_5	B, E



Induced Graph

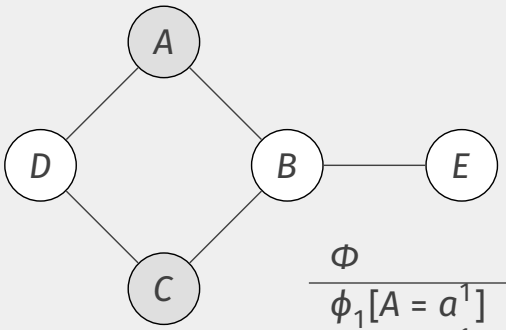
Φ	Scope	Available
$\phi_1[A = a^1]$	B	✗
$\phi_2[C = c^1]$	B	✗
$\phi_3[C = c^1]$	D	✓
$\phi_4[A = a^1]$	D	✓
ϕ_5	B, E	✗

Graphical Separation

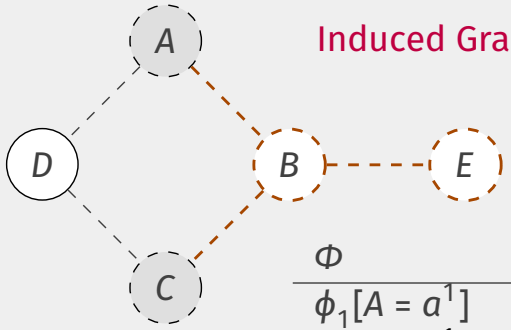
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$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
$\phi_3[C = c^1]$	D
$\phi_4[A = a^1]$	D
ϕ_5	B, E



Induced Graph

Φ	Scope	Available
$\phi_1[A = a^1]$	B	✗
$\phi_2[C = c^1]$	B	✗
$\phi_3[C = c^1]$	D	✓
$\phi_4[A = a^1]$	D	✓
ϕ_5	B, E	✗

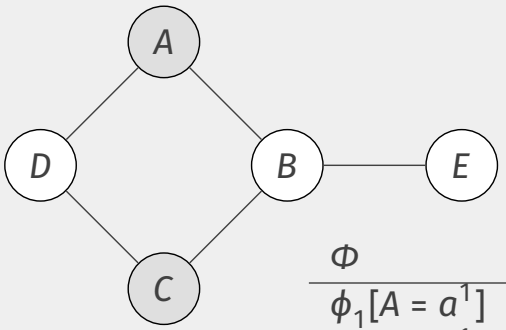
$$\tilde{P}_\Phi(D, A = a^1, C = c^1) = \phi_3[c^1](D)\phi_4[a^1](D)$$

Graphical Separation

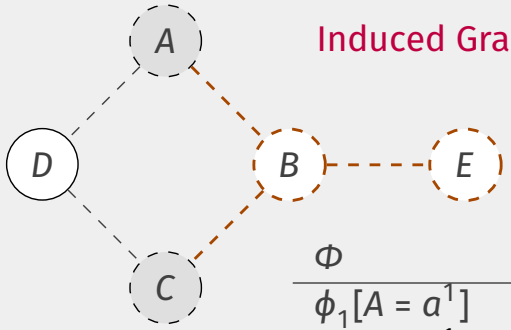
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Example: $P_\Phi(D|A = a^1, C = c^1)$ in *Misconception MN*



Φ	Scope
$\phi_1[A = a^1]$	B
$\phi_2[C = c^1]$	B
$\phi_3[C = c^1]$	D
$\phi_4[A = a^1]$	D
ϕ_5	B, E



Induced Graph

Φ	Scope	Available
$\phi_1[A = a^1]$	B	✗
$\phi_2[C = c^1]$	B	✗
$\phi_3[C = c^1]$	D	✓
$\phi_4[A = a^1]$	D	✓
ϕ_5	B, E	✗

$\tilde{P}_\Phi(D, A = a^1, C = c^1) = \phi_3[c^1](D)\phi_4[a^1](D)$. This induces the same $P_\Phi(D|A = a^1, C = c^1)$ as before because γ_3 is constant.

Each factor in Φ is used **once**.

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If the largest factor has N rvs in scope, each with cardinality K, then the **complexity of VE is proportional to** K^N.

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Like naive sum-product, worst-case VE is intractable. But most models have **sparse graph structure**, making VE often better.

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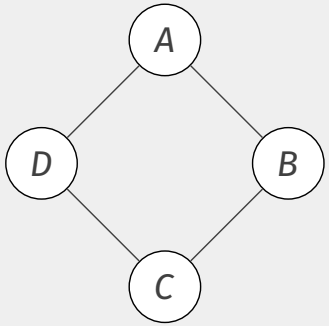
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Like naive sum-product, worst-case VE is intractable. But most models have **sparse graph structure**, making VE often better.

The **largest factor in VE depends on the order of elimination**. But, there's no polynomial-time algorithm to find an optimal ordering.

Exercise: normalisation constant

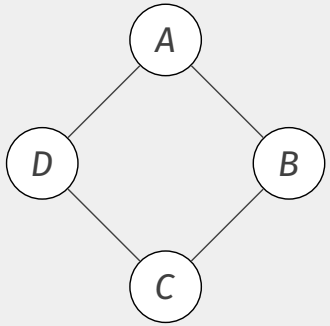
Obtain the normaliser for *Misconception* via VE and compare its computational complexity to the naive sum-product approach.



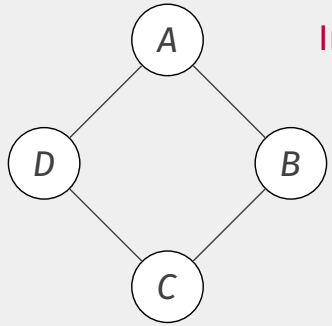
Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
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ϕ_4	D, A

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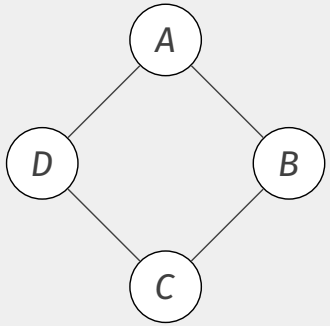
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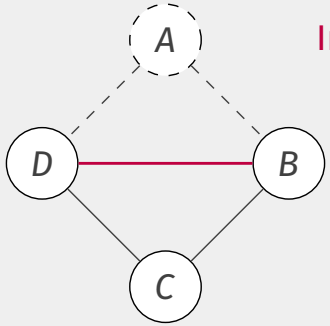
Induced Graph

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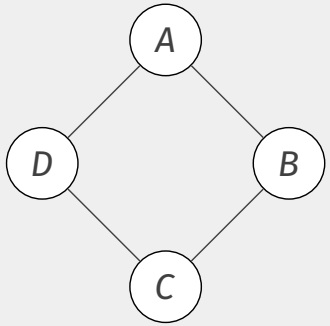


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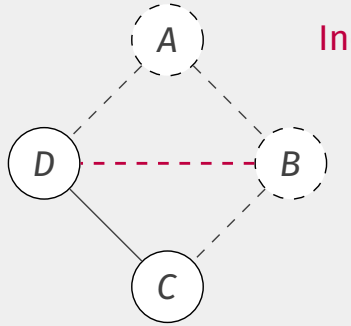
Elim.	Available	Relevant	Intermediate	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D) = \sum_A \rho_1(A, B, D)$

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ϕ_1	A, B
ϕ_2	B, C
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ϕ_4	D, A

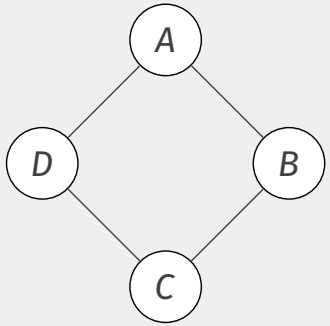


Induced Graph

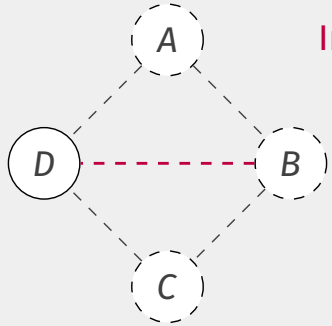
Elim.	Available	Relevant	Intermediate	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D) = \sum_A \rho_1(A, B, D)$
B	ϕ_2, ϕ_3, τ_1	ϕ_2, τ_1	$\rho_2(B, C, D)$	$\tau_2(C, D) = \sum_B \rho_2(B, C, D)$

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Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
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ϕ_4	D, A

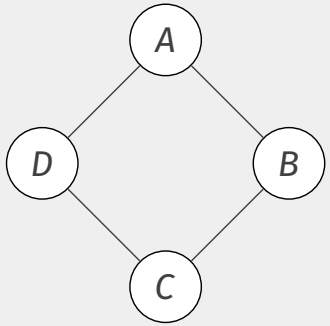


Induced Graph

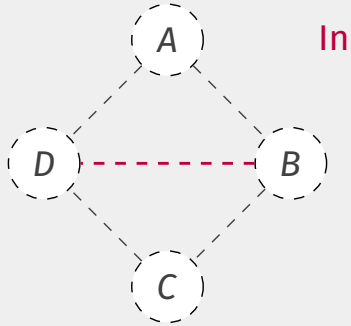
Elim.	Available	Relevant	Intermediate	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D) = \sum_A \rho_1(A, B, D)$
B	ϕ_2, ϕ_3, τ_1	ϕ_2, τ_1	$\rho_2(B, C, D)$	$\tau_2(C, D) = \sum_B \rho_2(B, C, D)$
C	ϕ_3, τ_2	ϕ_3, τ_2	$\rho_3(C, D)$	$\tau_3(D) = \sum_C \rho_3(C, D)$

Exercise: normalisation constant

Obtain the normaliser for *Misconception* via VE and compare its computational complexity to the naive sum-product approach.



Φ	Scope
ϕ_1	A, B
ϕ_2	B, C
ϕ_3	C, D
ϕ_4	D, A

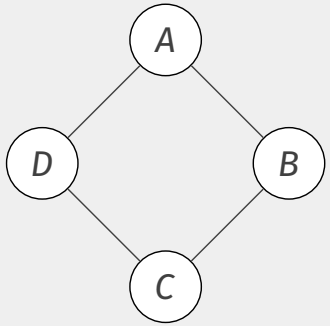


Induced Graph

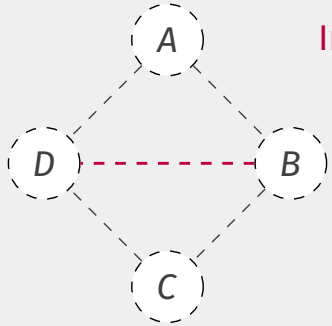
Elim.	Available	Relevant	Intermediate	New factor
A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D) = \sum_A \rho_1(A, B, D)$
B	ϕ_2, ϕ_3, τ_1	ϕ_2, τ_1	$\rho_2(B, C, D)$	$\tau_2(C, D) = \sum_B \rho_2(B, C, D)$
C	ϕ_3, τ_2	ϕ_3, τ_2	$\rho_3(C, D)$	$\tau_3(D) = \sum_C \rho_3(C, D)$
D	τ_3	τ_3	$\tau_3(D)$	$\gamma_4 = \sum_D \tau_3(D)$

Exercise: normalisation constant

Obtain the normaliser for *Misconception* via VE and compare its computational complexity to the naive sum-product approach.



Φ	Scope
ϕ_1	A, B
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ϕ_3	C, D
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Induced Graph

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A	$\phi_1, \phi_2, \phi_3, \phi_4$	ϕ_1, ϕ_4	$\rho_1(A, B, D)$	$\tau_1(B, D) = \sum_A \rho_1(A, B, D)$
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C	ϕ_3, τ_2	ϕ_3, τ_2	$\rho_3(C, D)$	$\tau_3(D) = \sum_C \rho_3(C, D)$
D	τ_3	τ_3	$\tau_3(D)$	$\gamma_4 = \sum_D \tau_3(D)$

$Z = \sum_{A,B,C,D} \tilde{P}_\Phi(A, B, C, D) = \gamma_4$. A naive approach to this sum requires the factor product of all 4 factors in Φ, whose table grows proportional to 2^4, while VE never creates a table larger than 2^3.

Summary

- Solving marginals is the key to complex probability queries (with or without evidence).
- Naive sum-product builds the largest possible factor—the (unnormalised) joint distribution—then factor marginalisation is safe to use. This algorithm is intractable for all but the smallest PGMs.
- With evidence, we can optimise naive sum-product a bit by reducing factors and by exploiting graphical separation, but a better solution needs to exploit more of the graphical structure (esp. neighbourhood).
- VE marginalises ('eliminates') one rv at a time in a given order. By taking products involving only the necessary factors at each step.
- VE achieves efficient exact inference, so long as the graph structure is sparse (which is true of many useful models).

What Next?

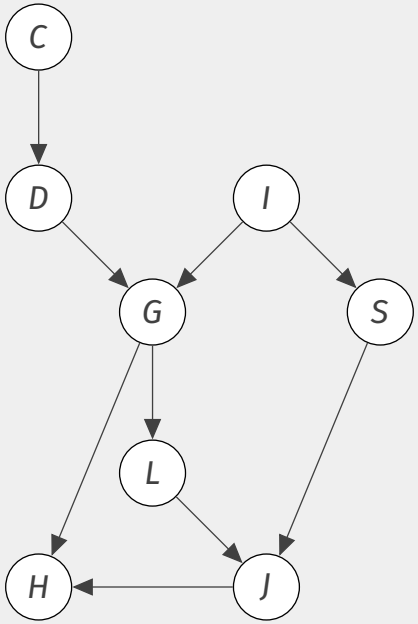
LC3: VE in code.

HC3b: elimination ordering and max-product inference.

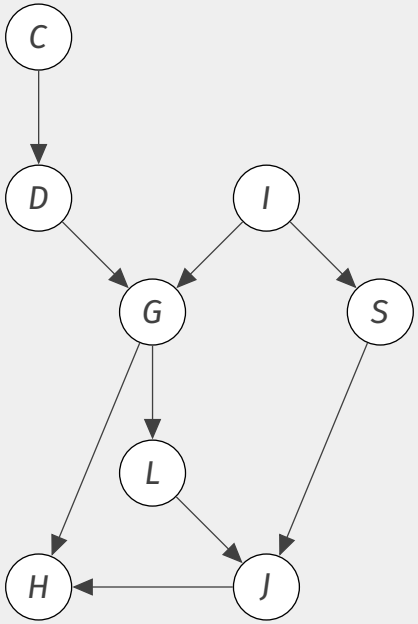
WC3: exercises.

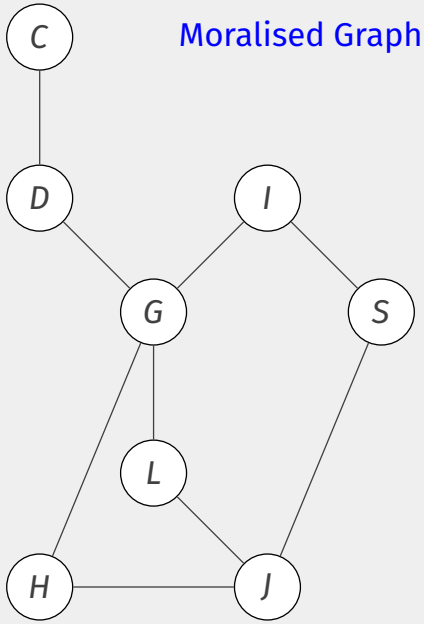
Elimination Ordering

Let's run VE to express $P(J)$ and let's start with eliminating G



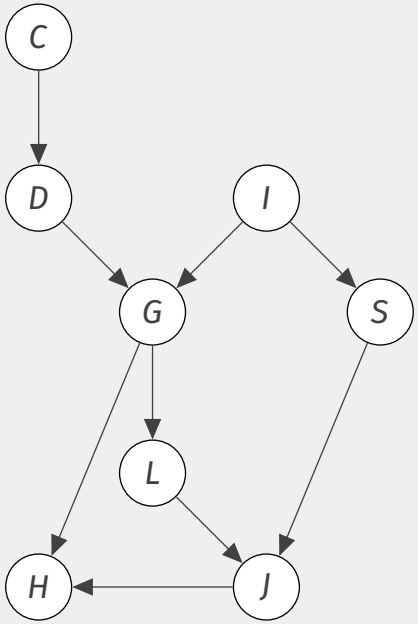
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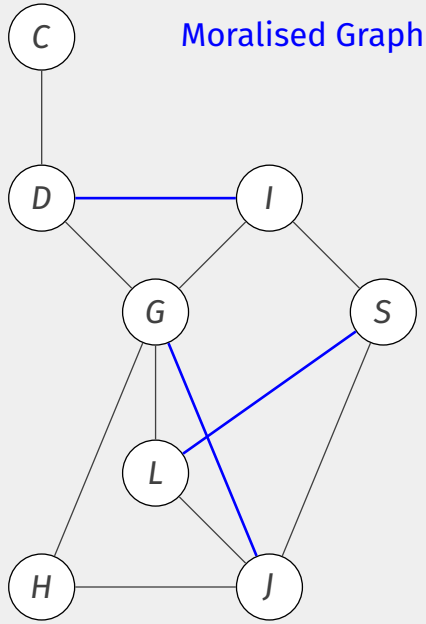




Moralised Graph

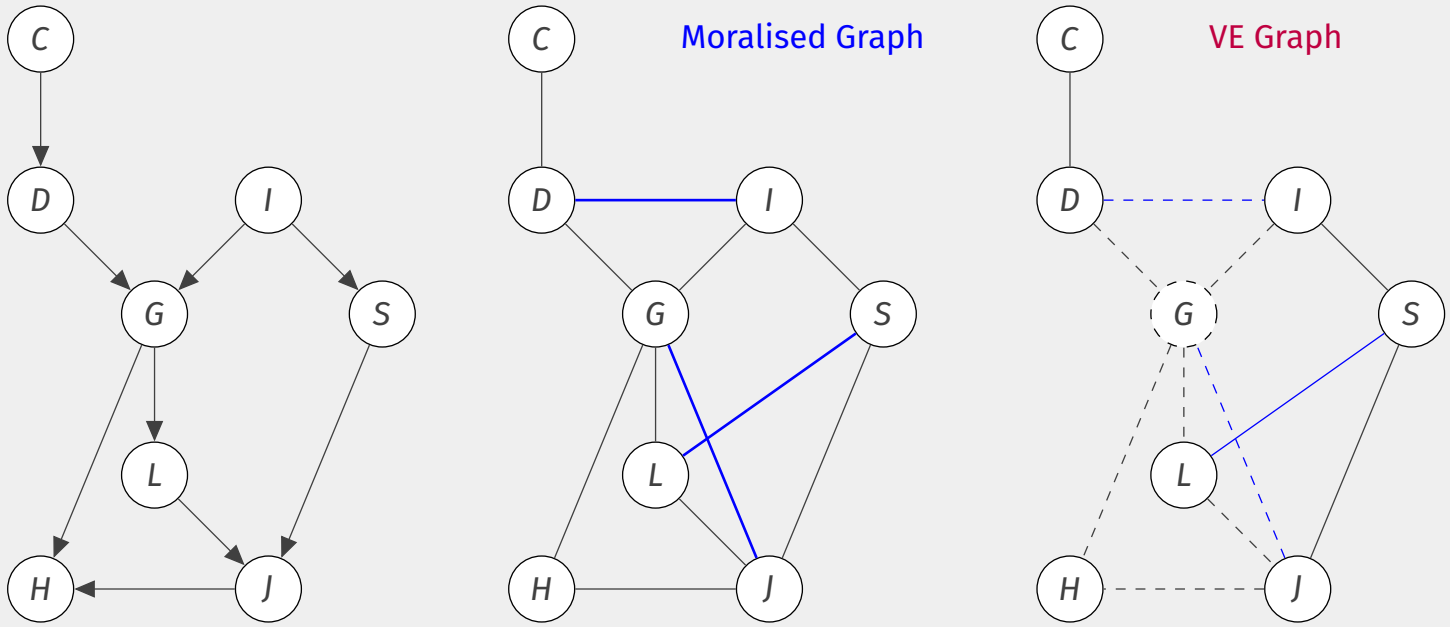
Let's run VE to express $P(J)$ and let's start with eliminating G





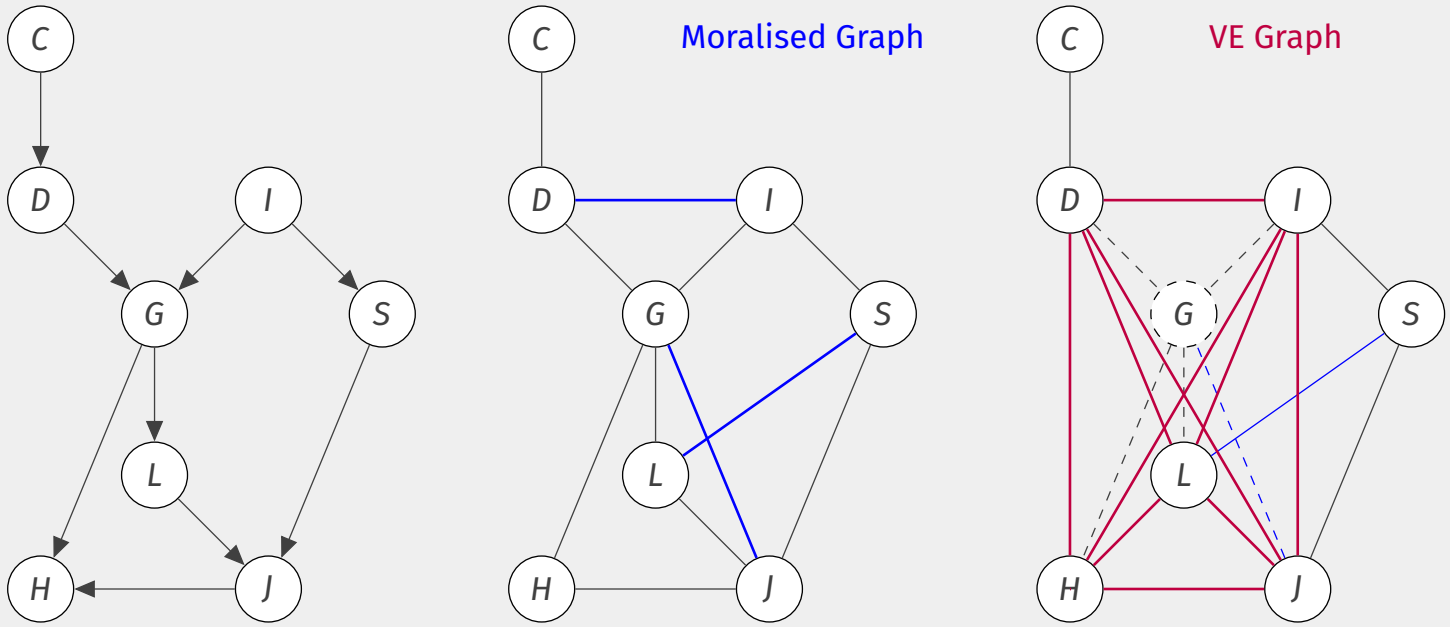
Moralised Graph

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All cliques involving G are dashed. Some or all of them will have factors in Φ, we should identify those factors and take their product.

Let's run VE to express $P(J)$ and let's start with eliminating G



All cliques involving G are dashed. Some or all of them will have factors in Φ, we should identify those factors and take their product.

This creates a **large factor** with scope D, G, H, I, J, L and cardinality 3×2^5.
In ex. 9.1, the largest factor had scope G, J, L, S and cardinality 3×2^3

Optimal Ordering

An optimal ordering is one for which the largest intermediate factor is as small as possible.

Unfortunately, finding such an order is intractable. [Section 9.4.3]

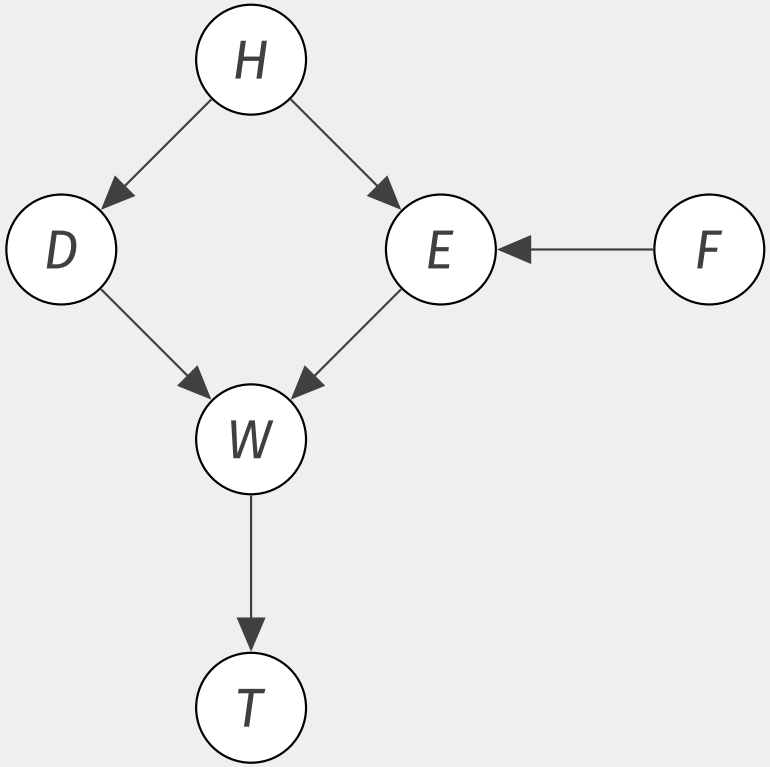
We resort to some useful heuristics: [Section 9.4.3.2]

- Min-neighbours
- Min-weight
- Min-fill
- Weighted-min-fill

These work surprisingly well in practice (min-fill and weighted-min-fill tend to work better on more problems).

Min-Neighbours

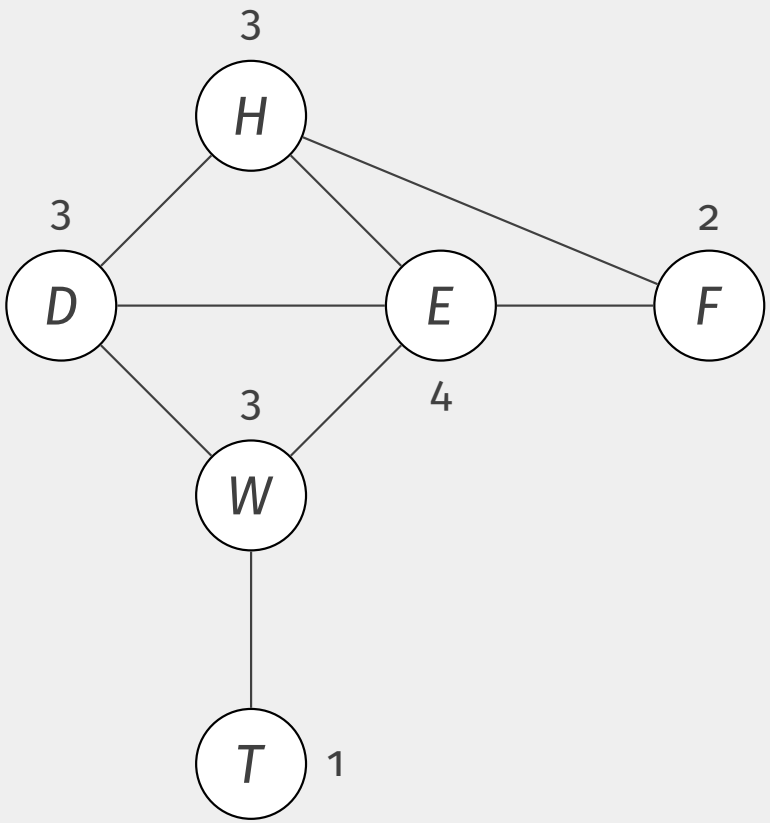
The cost of a node is the number of neighbours it has.



Moralise the BN and count neighbours.
Break ties lexicographically:

Min-Neighbours

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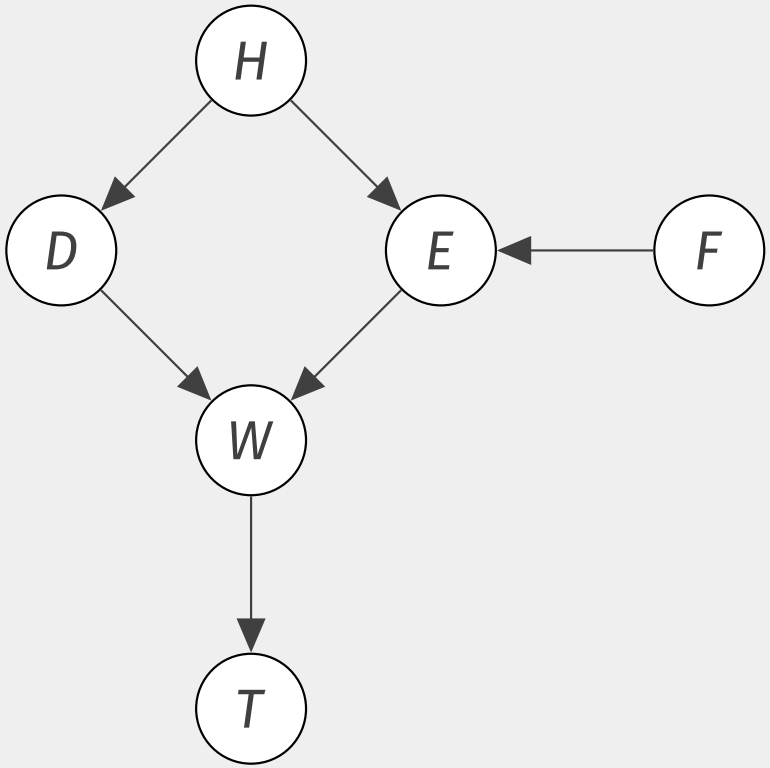


Moralise the BN and count neighbours.

Break ties lexicographically: T, F, D, H, W, E.

Min-Weight

The cost of a node is the product of *weights* (domain cardinality) of its neighbours. All rvs are binary except W which has cardinality 3.

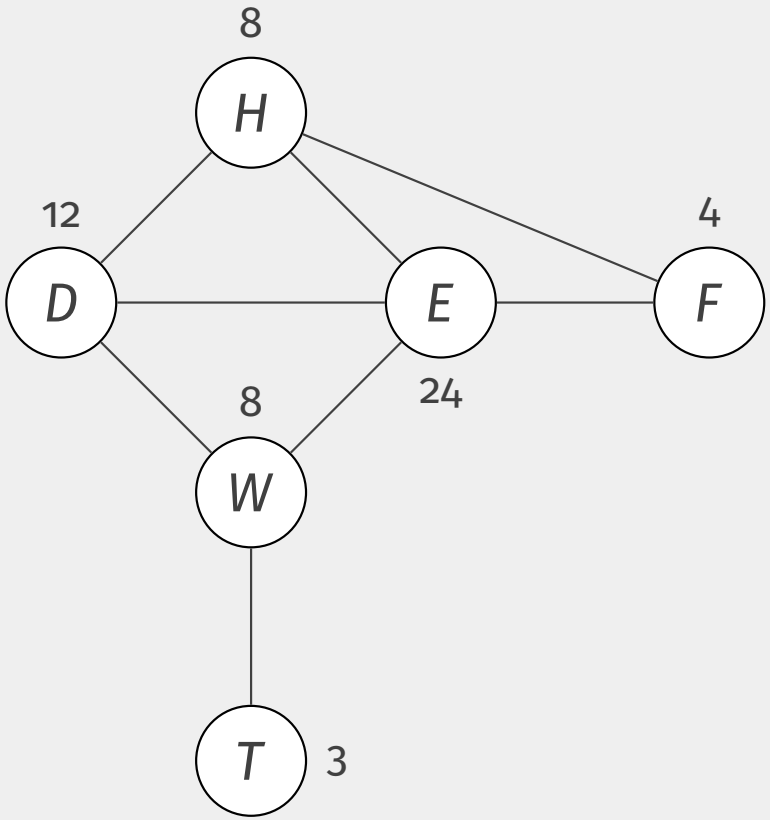


Moralise the BN and quantify weight.

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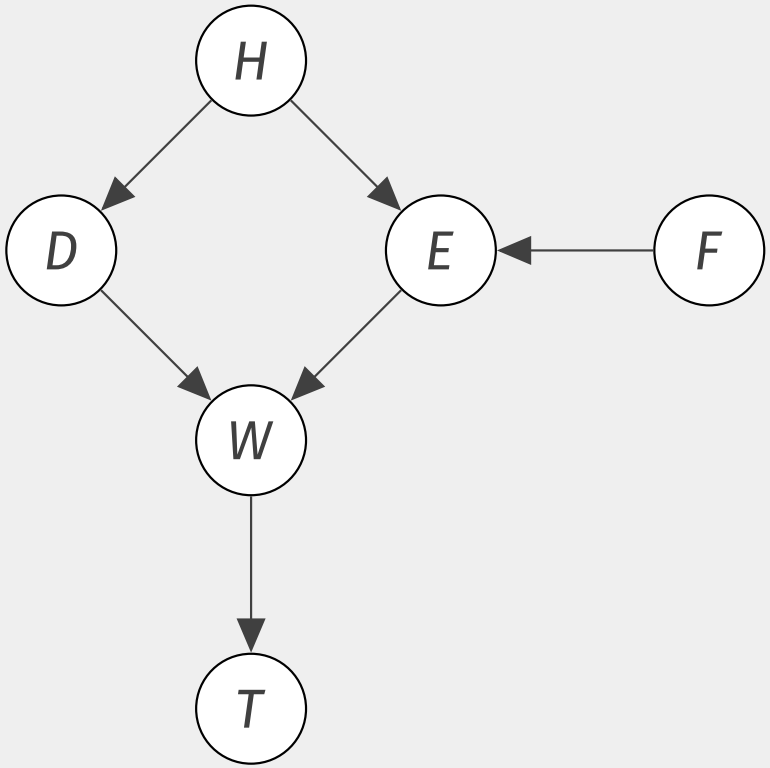


Moralise the BN and quantify weight.

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Min-Fill

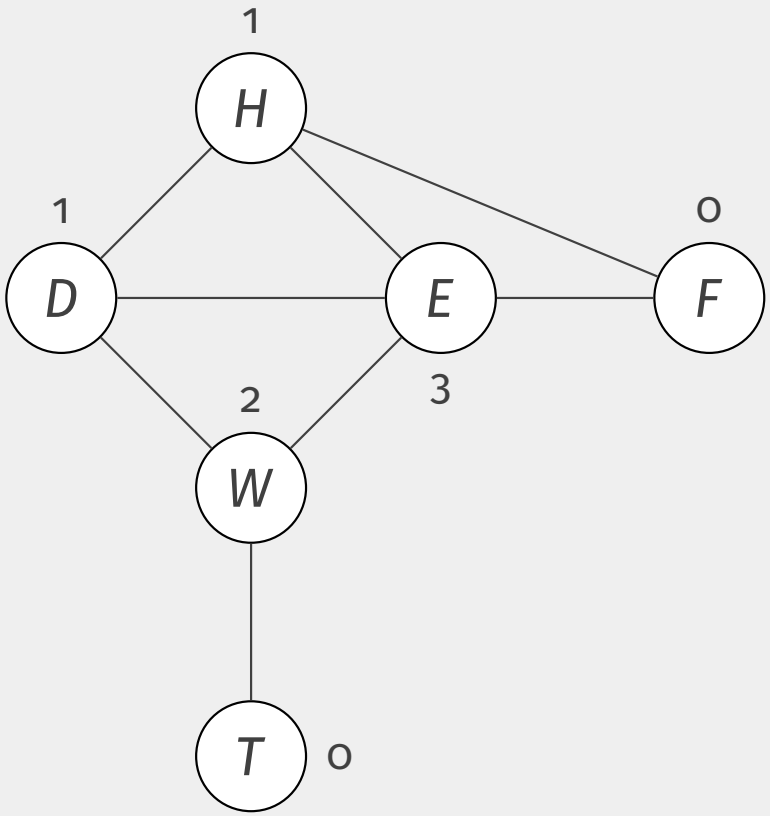
The cost of a node is the number of edges that need to be *added* to the graph due to its elimination.



Moralise the BN and count fill edges.
Break ties lexicographically:

Min-Fill

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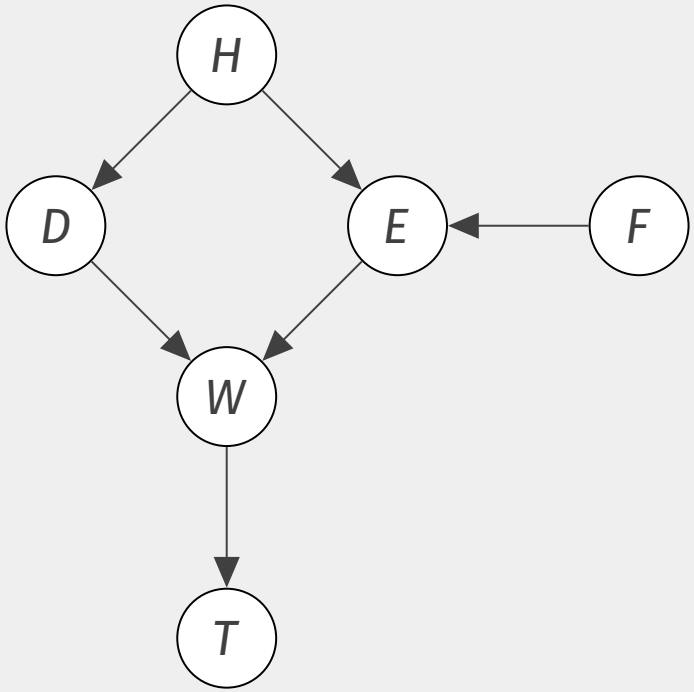


Moralise the BN and count fill edges.

Break ties lexicographically: F, T, D, H, W, E

Weighted-Min-Fill

The cost of a node is the sum of weights of the edges that need to be *added* to the graph due to its elimination, where weight of an edge is the product of weights of its vertices. All rvs are binary except W which has cardinality 3.

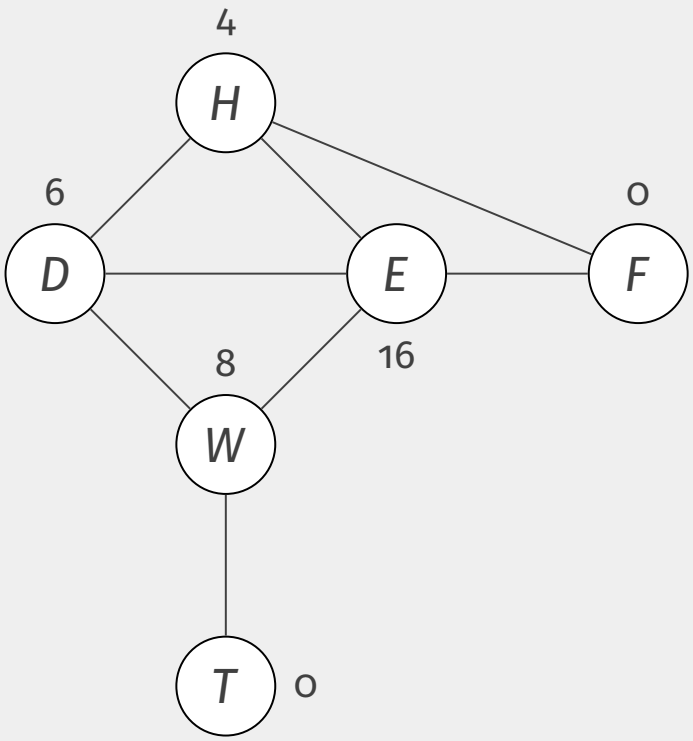


Moralise the BN and quantify total weight of fill edges.

Break ties lexicographically:

Weighted-Min-Fill

The cost of a node is the sum of weights of the edges that need to be *added* to the graph due to its elimination, where weight of an edge is the product of weights of its vertices. All rvs are binary except W which has cardinality 3.



Moralise the BN and quantify total weight of fill edges.

Break ties lexicographically: F, T, H, D, W, E

Induced Graph and Search for Elimination Ordering

Suppose we have 4 heuristics (such as those just presented), we can simulate their corresponding induced graphs in VE (that is, we draw the induced graph without actually taking any factor product).

Then we can find out which heuristic leads to the most efficient VE run: the one whose induced graph requires the least prohibitive factor products.

Induced Graph and Search for Elimination Ordering

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Then we can find out which heuristic leads to the most efficient VE run: the one whose induced graph requires the least prohibitive factor products.

We may recompute the heuristics dynamically as we eliminate variables, since this can affect the ordering. But, for simplicity, in this course we are establishing the order once from the moralised graph, before running VE.

Summary

- The most efficient VE run uses the elimination order which creates the smallest intermediate factor or smallest ‘maximal clique’.
- Optimal orderings are intractable, but simple heuristics work rather well in practice.

Max-Product (or MAP) Inference

Yet another inference task!

We've seen many types of inference tasks, and tools to approach them:

- marginals $P_\Phi(\mathbf{Y}) = \sum_{\mathbf{M}} P_\Phi(\mathbf{M}, \mathbf{Y})$
- conditionals $P_\Phi(\mathbf{Y} | \mathbf{E} = \mathbf{e}) = \frac{P_\Phi(\mathbf{Y}, \mathbf{E}=\mathbf{e})}{P_\Phi(\mathbf{E}=\mathbf{e})}$
- a combination of both $P_\Phi(\mathbf{Q} | \mathbf{E} = \mathbf{e}) = \frac{\sum_{\mathbf{M}} P_\Phi(\mathbf{Q}, \mathbf{M}, \mathbf{E}=\mathbf{e})}{P_\Phi(\mathbf{E}=\mathbf{e})}$

And now we cover a new one: **maximum a-posteriori** (MAP) inference is the task of finding the mode of the distribution.

$$\mathbf{x}^* = \operatorname*{argmax}_{\mathbf{x} \in \mathrm{Val}(\mathbf{X})} P_\Phi(\mathbf{X})$$

This is often used as a decision rule in ML tasks.

Simplest

$$\mathbf{x}^* = \operatorname*{argmax}_{\mathbf{x} \in \text{Val}(\mathbf{X})} P_\Phi(\mathbf{X})$$

Simplest

$$\mathbf{x}^* = \underset{\mathbf{x} \in \text{Val}(\mathbf{X})}{\text{argmax}} \; P_\Phi(\mathbf{X}) = \underset{\mathbf{x} \in \text{Val}(\mathbf{X})}{\text{argmax}} \; \tilde{P}_\Phi(\mathbf{X})$$

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$$\mathbf{x}^* = \underset{\mathbf{x} \in \text{Val}(\mathbf{X})}{\text{argmax}} \; P_\Phi(\mathbf{X}) = \underset{\mathbf{x} \in \text{Val}(\mathbf{X})}{\text{argmax}} \; \tilde{P}_\Phi(\mathbf{X})$$

With evidence $\mathbf{E} = \mathbf{e}$ where $\mathbf{Z} = \mathbf{X} \setminus \mathbf{X}$

$$\mathbf{z}^* = \underset{\mathbf{z} \in \text{Val}(\mathbf{Z})}{\text{argmax}} \; P_\Phi(\mathbf{Z} | \mathbf{E} = \mathbf{e})$$

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Marginal query $\mathbf{Q}$ where $\mathbf{M} = \mathbf{X} \setminus \mathbf{Q}$:

$$\mathbf{q}^* = \underset{\mathbf{q} \in \text{Val}(\mathbf{Q})}{\text{argmax}} \sum_{\mathbf{M}} P_\Phi(\mathbf{M}, \mathbf{Q})$$

Simplest

$$\mathbf{x}^* = \underset{\mathbf{x} \in \text{Val}(\mathbf{X})}{\text{argmax}} \; P_\Phi(\mathbf{X}) = \underset{\mathbf{x} \in \text{Val}(\mathbf{X})}{\text{argmax}} \; \tilde{P}_\Phi(\mathbf{X})$$

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Marginal query $\mathbf{Q}$ with $\mathbf{E} = \mathbf{e}$ where $\mathbf{Q} \cap \mathbf{E} = \emptyset$ and $\mathbf{M} = \mathbf{X} \setminus (\mathbf{Q} \cup \mathbf{E})$

$$\mathbf{q}^* = \operatorname*{argmax}_{\mathbf{q} \in \text{Val}(\mathbf{Q})} \sum_{\mathbf{M}} P_\Phi(\mathbf{M}, \mathbf{Q} | \mathbf{E} = \mathbf{e})$$

Simplest

$$\mathbf{x}^* = \operatorname*{argmax}_{\mathbf{x} \in \text{Val}(\mathbf{X})} P_\Phi(\mathbf{X}) = \operatorname*{argmax}_{\mathbf{x} \in \text{Val}(\mathbf{X})} \tilde{P}_\Phi(\mathbf{X})$$

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A	ϕ
a^0	3
a^1	5

$\operatorname{argmax}_A \phi(A)$	$\max_A \phi(A)$

A	ϕ
a^0	3
a^1	5

$\operatorname{argmax}_A \phi(A)$	$\max_A \phi(A)$
a^1	5

Short Digression: Factor Maximisation Example

B	C	ϕ
b^0	c^0	100
b^0	c^1	1
b^1	c^0	2
b^1	c^1	10

B	$\delta(B) = \text{argmax}_C \ \phi(B, C)$
b^0	
b^1	

B	$\psi(B) = \max_C \phi(B, C)$
b^0	
b^1	

Short Digression: Factor Maximisation Example

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There is a relationship between factor maximisation $\psi(B)$ and the argmax function $\delta(B)$, namely, $\psi(B) = \phi(B, C = \delta(B))$.

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b^0	100
b^1	10

There is a relationship between factor maximisation $\psi(B)$ and the argmax function $\delta(B)$, namely, $\psi(B) = \phi(B, C = \delta(B))$.

Note that the argmax function is not a factor in the standard sense (its co-domain is not $\mathbb{R}_{\geq 0}$, it is $\text{Val}(B)$).

Factor Maximisation [Definition 13.2]

We have a set of rvs $\mathbf{X}$ and an rv $Y \notin \mathbf{X}$ and a factor $\phi(\mathbf{X}, Y)$.

The **factor maximisation** of Y in ϕ is the factor

$$\psi(\mathbf{X}) = \max_{y \in \text{Val}(Y)} \phi(\mathbf{X}, Y = y)$$

whose scope is $\text{Scope}[\psi] = \mathbf{X}$.

Factor Maximisation [Definition 13.2]

We have a set of rvs $\mathbf{X}$ and an rv $Y \notin \mathbf{X}$ and a factor $\phi(\mathbf{X}, Y)$.

The **factor maximisation** of Y in ϕ is the factor

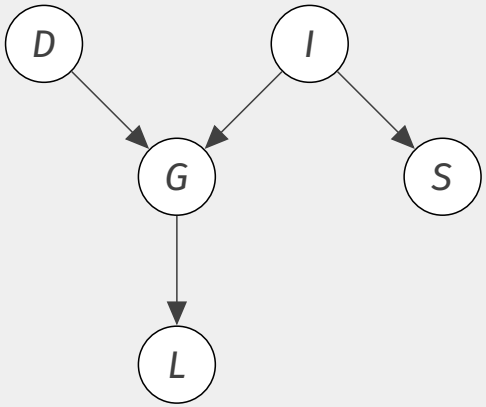
$$\psi(\mathbf{X}) = \max_{y \in \text{Val}(Y)} \phi(\mathbf{X}, Y = y)$$

whose scope is $\text{Scope}[\psi] = \mathbf{X}$.

Useful properties: for $Y \notin \mathbf{X}$ and $Y \notin \mathbf{Z}$

- $\max_Y \ \phi_1(\mathbf{X}) \times \phi_2(\mathbf{Z}, Y) = \phi_1(\mathbf{X}) \times \max_Y \ \phi_2(\mathbf{Z}, Y)$
- $\max_Y \ \phi_1(\mathbf{X}) + \phi_2(\mathbf{Z}, Y) = \phi_1(\mathbf{X}) + \max_Y \ \phi_2(\mathbf{Z}, Y)$

Let's look for the mode of the *Student example*.



Factor	Scope
ϕ_D	D
ϕ_I	I
ϕ_S	I, S
ϕ_G	D, I, G
ϕ_L	G, L

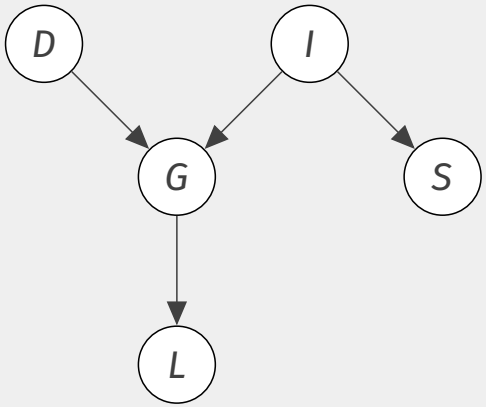
We are looking to solve

$$\operatorname*{argmax}_{D,I,S,G,L} P(D, I, S, G, L)$$

Let's start with the maximisation problem

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G, D, I)\phi_S(S, I)\phi_L(L, G)$$

Let's look for the mode of the *Student example*.



Factor	Scope
ϕ_D	D
ϕ_I	I
ϕ_S	I, S
ϕ_G	D, I, G
ϕ_L	G, L

We are looking to solve

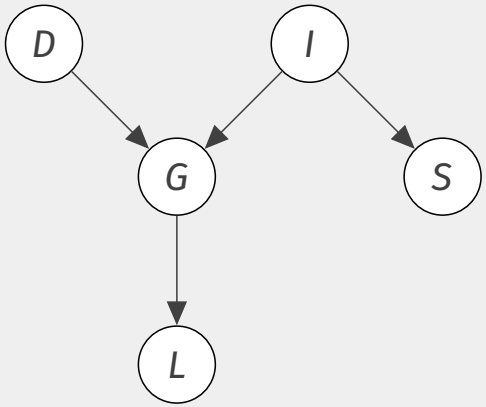
$$\operatorname*{argmax}_{D,I,S,G,L} P(D, I, S, G, L)$$

Let's start with the maximisation problem

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G, D, I)\phi_S(S, I)\phi_L(L, G)$$

Finding the max is what we call a **max-product** inference problem. Finding the argmax is what we call a **decoding** problem.

Variable Elimination for Max-Product Inference [Section 13.2]



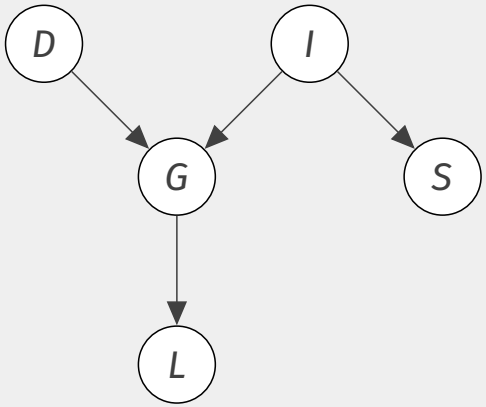
Factor	Scope
ϕ_D	D
ϕ_I	I
ϕ_S	I, S
ϕ_G	D, I, G
ϕ_L	G, L

We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

As we learnt in the case of sum-product inference, we benefit from rearranging the maximisation and the products in ways that exploit the graph structure to avoid constructing large factors.

Variable Elimination for Max-Product Inference [Section 13.2]



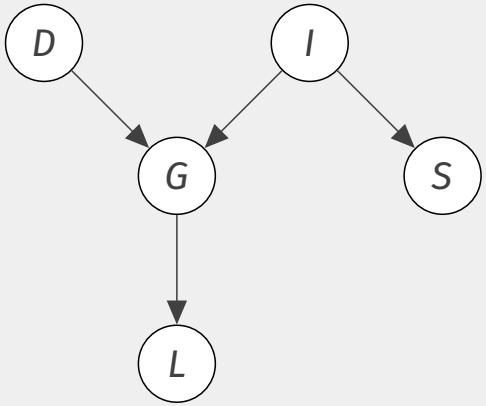
Factor	Scope
ϕ_D	D
ϕ_I	I
ϕ_S	I, S
ϕ_G	D, I, G
ϕ_L	G, L

We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

As we learnt in the case of sum-product inference, we benefit from rearranging the maximisation and the products in ways that exploit the graph structure to avoid constructing large factors.

Next, let's approach the problem one rv at a time, in some order such as L, S, D, I, G, so that we can use what we learnt about *factor maximisation*.

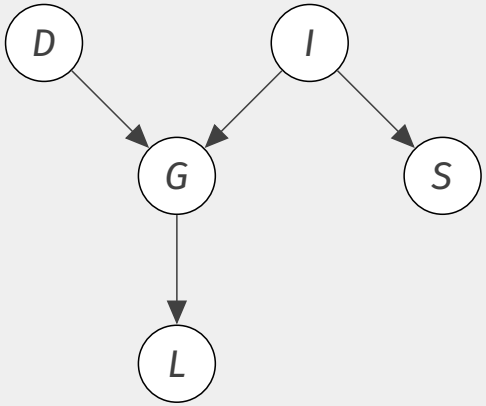


We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate *L*:

$$\max_{L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$



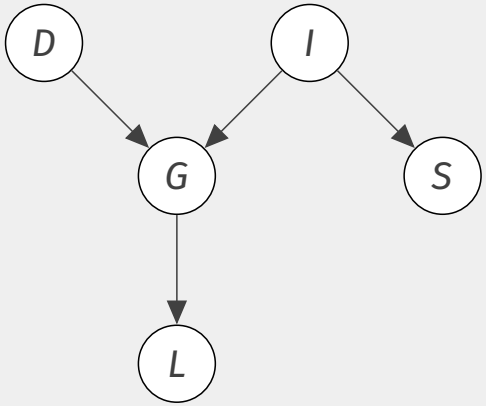
We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate *L*:

$$\max_{L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

$$= \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I) \underbrace{\max_{L} \phi_L(L,G)}_{\tau_1(G)}$$



We are looking to solve a factor maximisation for a large factor product:

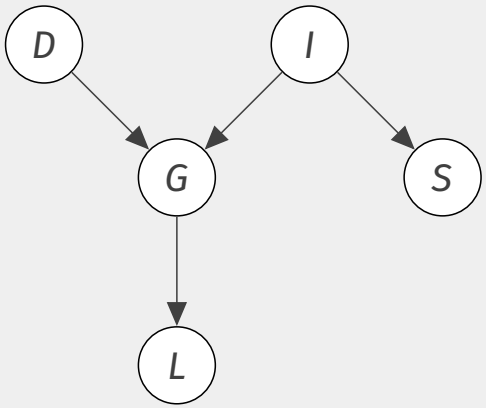
$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate L:

$$\max_L \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

$$= \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I) \underbrace{\max_L \phi_L(L,G)}_{\tau_1(G)}$$

$$= \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\tau_1(G)$$

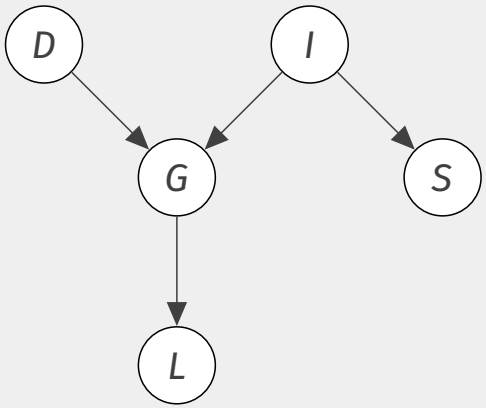


We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate S:

$$\max_{S} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\tau_1(G)$$



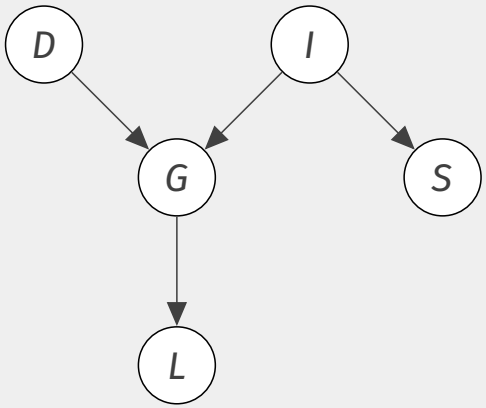
We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate S:

$$\max_S \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\tau_1(G)$$

$$= \phi_D(D)\phi_I(I)\phi_G(G,D,I)\tau_1(G) \underbrace{\max_S \phi_S(S,I)}_{\tau_2(I)}$$



We are looking to solve a factor maximisation for a large factor product:

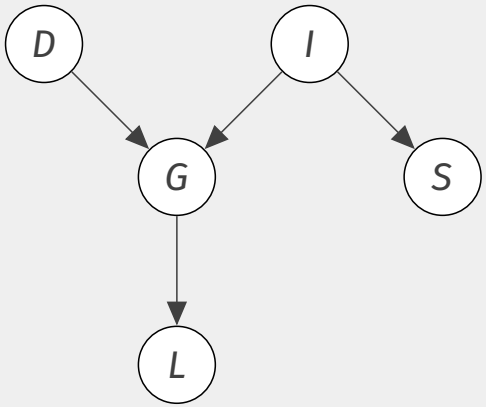
$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate S:

$$\max_S \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\tau_1(G)$$

$$= \phi_D(D)\phi_I(I)\phi_G(G,D,I)\tau_1(G) \underbrace{\max_S \phi_S(S,I)}_{\tau_2(I)}$$

$$= \phi_D(D)\phi_I(I)\phi_G(G,D,I)\tau_1(G)\tau_2(I)$$

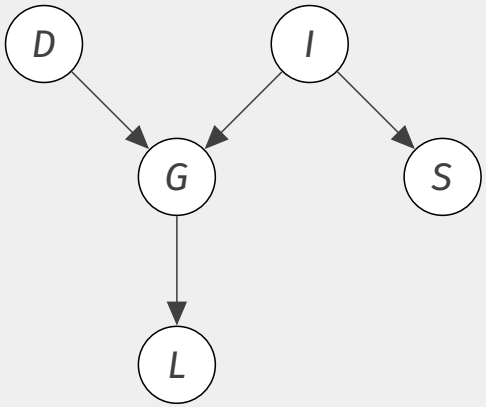


We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate *D*:

$$\max_{L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\tau_1(G)\tau_2(I)$$



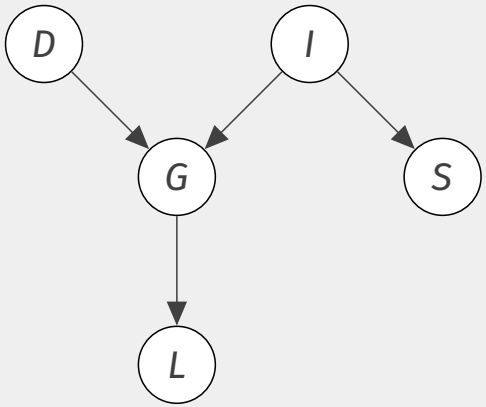
We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate *D*:

$$\max_L \phi_D(D)\phi_I(I)\phi_G(G,D,I)\tau_1(G)\tau_2(I)$$

$$= \phi_I(I)\tau_1(G)\tau_2(I) \underbrace{\max_L \overbrace{\phi_D(D)\phi_G(G,D,I)}^{\rho_3(G,D,I)}}_{\tau_3(G,I)}$$



We are looking to solve a factor maximisation for a large factor product:

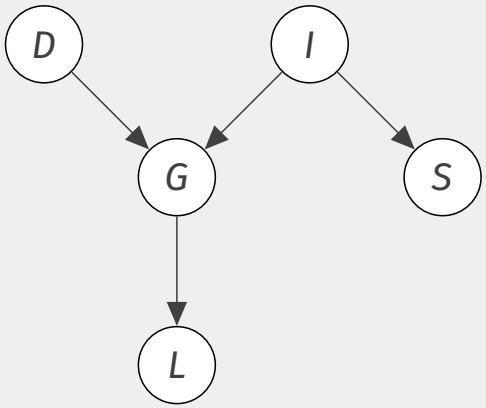
$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate *D*:

$$\max_L \phi_D(D)\phi_I(I)\phi_G(G,D,I)\tau_1(G)\tau_2(I)$$

$$= \phi_I(I)\tau_1(G)\tau_2(I) \underbrace{\max_L \overbrace{\phi_D(D)\phi_G(G,D,I)}^{\rho_3(G,D,I)}}_{\tau_3(G,I)}$$

$$= \phi_I(I)\tau_1(G)\tau_2(I)\tau_3(G,I)$$

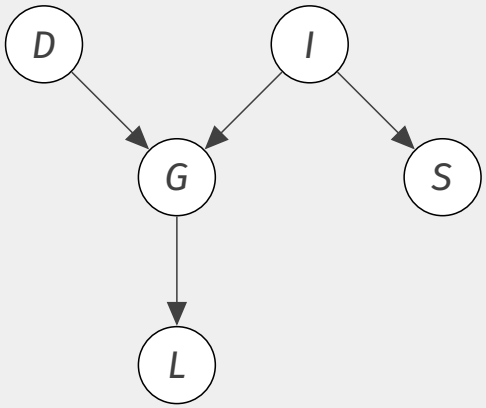


We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate I:

$$\max_I \phi_I(I)\tau_1(G)\tau_2(I)\tau_3(G,I)$$



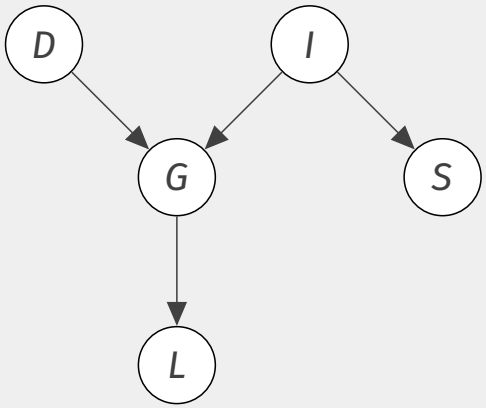
Eliminate I:

We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

$$\max_I \phi_I(I)\tau_1(G)\tau_2(I)\tau_3(G,I)$$

$$= \tau_1(G) \underbrace{\max_I \overbrace{\phi_I(I)\tau_2(I)\tau_3(G,I)}^{\rho_4(G,I)}}_{\tau_4(G)}$$



Eliminate I:

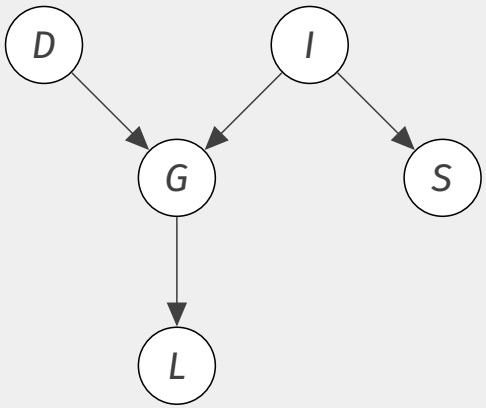
We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

$$\max_I \phi_I(I)\tau_1(G)\tau_2(I)\tau_3(G,I)$$

$$= \tau_1(G) \underbrace{\max_I \overbrace{\phi_I(I)\tau_2(I)\tau_3(G,I)}^{\rho_4(G,I)}}_{\tau_4(G)}$$

$$= \tau_1(G)\tau_4(G)$$

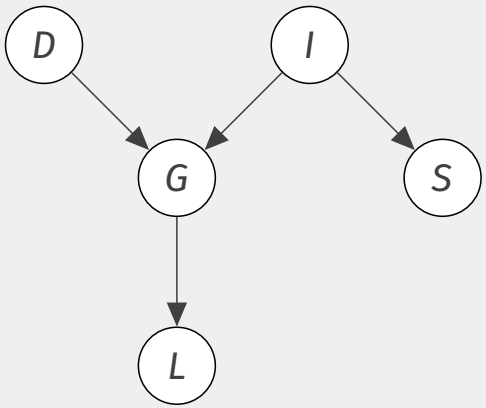


Eliminate G:

We are looking to solve a factor maximisation for a large factor product:

$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

$$\max_{G} \overbrace{\tau_1(G)\tau_4(G)}^{\rho_5(G)}$$



We are looking to solve a factor maximisation for a large factor product:

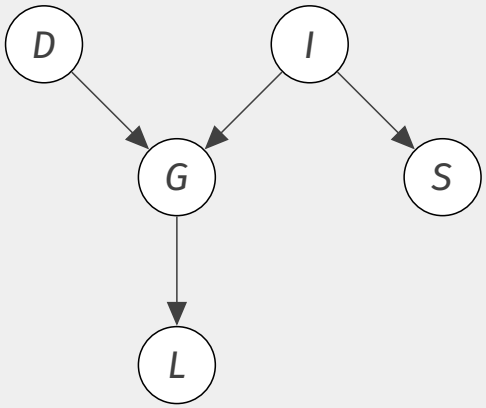
$$\max_{D,I,S,G,L} \phi_D(D)\phi_I(I)\phi_G(G,D,I)\phi_S(S,I)\phi_L(L,G)$$

Eliminate G:

$$\max_{G} \overbrace{\tau_1(G)\tau_4(G)}^{\rho_5(G)}$$

$$= \gamma_5$$

Here we are: γ_5 **is** the (unnormalised) probability of the mode.

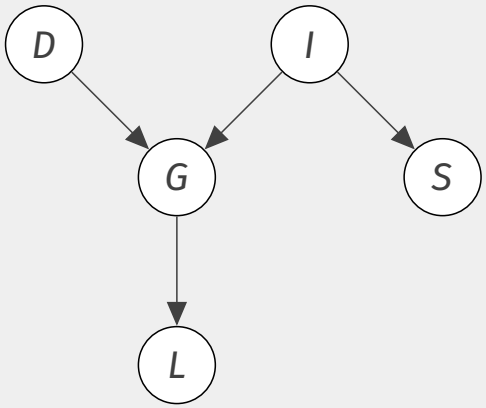


Start with the joint distribution, but let's write it using *factors*:

$$P(D, I, G, S, L)$$
$$= \phi_D(D)\phi_I(I)\phi_G(G, D, I)\phi_S(S, I)\phi_L(L, G)$$

Elim. X	**Available**	**Intermediate (prod. of relevant)**	**Final** $\max_X \rho$	**Traceback**
L	$\phi_D, \phi_I, \phi_S, \phi_G, \phi_L$	$\rho_1(G, L) = \phi_L(G, L)$	$\tau_1(G)$	

Table 13.1: a run of max-prod VE using a different ordering.

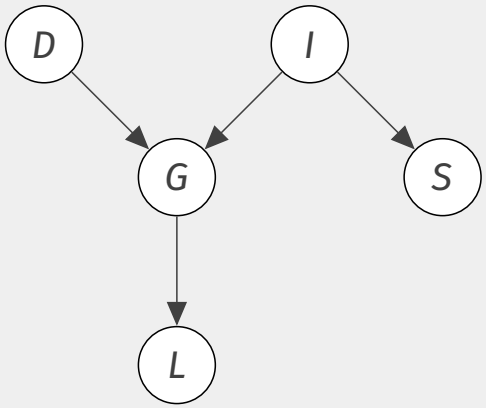


Start with the joint distribution, but let's write it using *factors*:

$$P(D, I, G, S, L)$$
$$= \phi_D(D)\phi_I(I)\phi_G(G, D, I)\phi_S(S, I)\phi_L(L, G)$$

Elim. X	**Available**	**Intermediate (prod. of relevant)**	**Final** $\max_X \rho$	**Traceback**
L	$\phi_D, \phi_I, \phi_S, \phi_G, \phi_L$	$\rho_1(G, L) = \phi_L(G, L)$	$\tau_1(G)$	
S	$\phi_D, \phi_I, \phi_S, \phi_G, \tau_1$	$\rho_2(S, I) = \phi_S(S, I)$	$\tau_2(I)$	

Table 13.1: a run of max-prod VE using a different ordering.

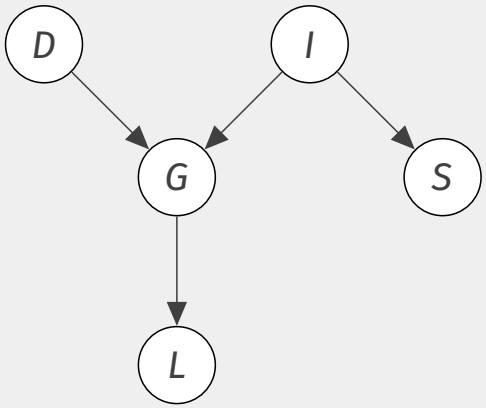


Start with the joint distribution, but let's write it using *factors*:

$$P(D, I, G, S, L)$$
$$= \phi_D(D)\phi_I(I)\phi_G(G, D, I)\phi_S(S, I)\phi_L(L, G)$$

Elim. X	**Available**	**Intermediate (prod. of relevant)**	**Final** $\max_X \rho$	**Traceback**
L	$\phi_D, \phi_I, \phi_S, \phi_G, \phi_L$	$\rho_1(G, L) = \phi_L(G, L)$	$\tau_1(G)$	
S	$\phi_D, \phi_I, \phi_S, \phi_G, \tau_1$	$\rho_2(S, I) = \phi_S(S, I)$	$\tau_2(I)$	
D	$\phi_D, \phi_I, \phi_G, \tau_1, \tau_2$	$\rho_3(G, D, I) = \phi_D(D)\phi_G(G, D, I)$	$\tau_3(G, I)$	

Table 13.1: a run of max-prod VE using a different ordering.

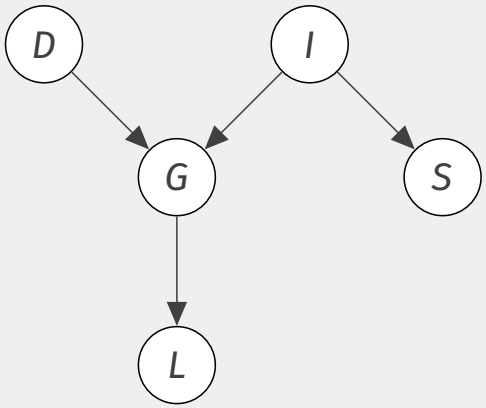


Start with the joint distribution, but let's write it using *factors*:

$$P(D, I, G, S, L)$$
$$= \phi_D(D)\phi_I(I)\phi_G(G, D, I)\phi_S(S, I)\phi_L(L, G)$$

Elim. X	**Available**	**Intermediate (prod. of relevant)**	**Final** $\max_X \rho$	**Traceback**
L	$\phi_D, \phi_I, \phi_S, \phi_G, \phi_L$	$\rho_1(G, L) = \phi_L(G, L)$	$\tau_1(G)$	
S	$\phi_D, \phi_I, \phi_S, \phi_G, \tau_1$	$\rho_2(S, I) = \phi_S(S, I)$	$\tau_2(I)$	
D	$\phi_D, \phi_I, \phi_G, \tau_1, \tau_2$	$\rho_3(G, D, I) = \phi_D(D)\phi_G(G, D, I)$	$\tau_3(G, I)$	
I	$\phi_I, \tau_1, \tau_2, \tau_3$	$\rho_4(G, I) = \phi_I(I)\tau_2(I)\tau_3(G, I)$	$\tau_4(G)$	

Table 13.1: a run of max-prod VE using a different ordering.

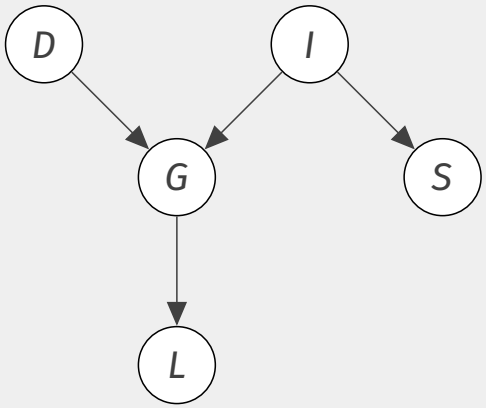


Start with the joint distribution, but let's write it using *factors*:

$$P(D, I, G, S, L) = \phi_D(D)\phi_I(I)\phi_G(G, D, I)\phi_S(S, I)\phi_L(L, G)$$

Elim. X	Available	Intermediate (prod. of relevant)	Final $\max_X \rho$	Traceback
L	$\phi_D, \phi_I, \phi_S, \phi_G, \phi_L$	$\rho_1(G, L) = \phi_L(G, L)$	$\tau_1(G)$	
S	$\phi_D, \phi_I, \phi_S, \phi_G, \tau_1$	$\rho_2(S, I) = \phi_S(S, I)$	$\tau_2(I)$	
D	$\phi_D, \phi_I, \phi_G, \tau_1, \tau_2$	$\rho_3(G, D, I) = \phi_D(D)\phi_G(G, D, I)$	$\tau_3(G, I)$	
I	$\phi_I, \tau_1, \tau_2, \tau_3$	$\rho_4(G, I) = \phi_I(I)\tau_2(I)\tau_3(G, I)$	$\tau_4(G)$	
G	τ_1, τ_4	$\rho_5(G) = \tau_1(G)\tau_4(G)$	γ_5	

Table 13.1: a run of max-prod VE using a different ordering.

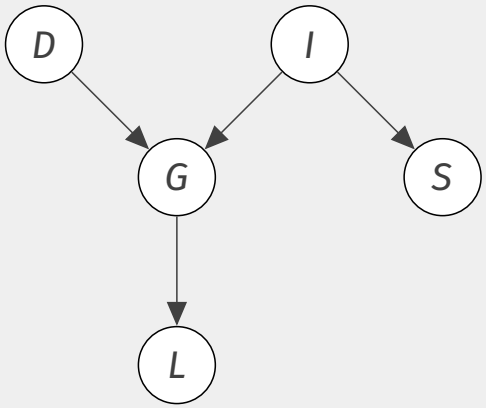


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Elim. X	**Available**	**Intermediate (prod. of relevant)**	**Final** $\max_X \rho$	**Traceback**
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D	$\phi_D, \phi_I, \phi_G, \tau_1, \tau_2$	$\rho_3(G, D, I) = \phi_D(D)\phi_G(G, D, I)$	$\tau_3(G, I)$	
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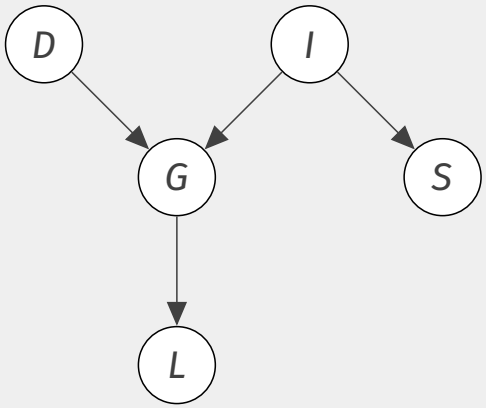


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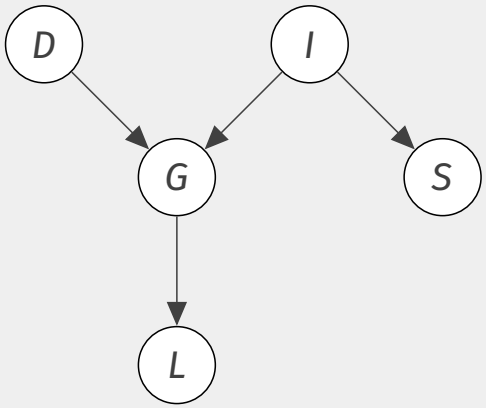


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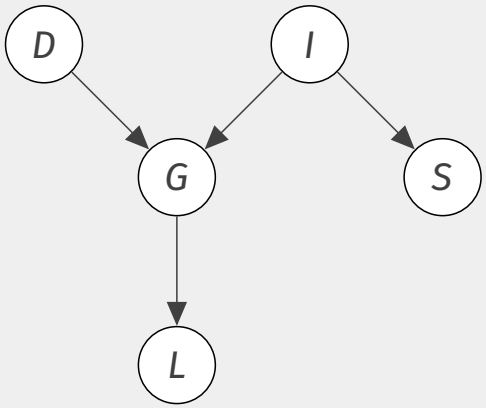


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S	$\phi_D, \phi_I, \phi_S, \phi_G, \tau_1$	$\rho_2(S, I) = \phi_S(S, I)$	$\tau_2(I)$	$s^* = \text{argmax}_S\, \rho_2(S, I = i^*)$
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1. gather all relevant factors—those for which Y is in the scope—removing them from Φ;
2. take the product of the relevant factors;
3. maximise Y out of the product;
4. add the resulting factor in the collection;
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Decoding. Traverse the trace in reverse constructing the argmax assignment $\mathbf{x}^*$. By construction, you will be able to reduce the rvs in the scope using the partial assignment $\mathbf{o}^*$ already built at each point. Hence, each time you solve $\operatorname{argmax}_Y \rho[\mathbf{o}^*](Y)$.

Max-Product VE for Other MAP Tasks

(1) With **evidence** $\text{argmax}_{\mathbf{z} \in \text{Val}(\mathbf{Z})} \tilde{P}_\Phi(\mathbf{Z}, \mathbf{E} = \mathbf{e})$

Reduce factors first, then eliminate $\mathbf{Z}$.

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(3) For a **marginal query with evidence**: (1) then (2).

Summary

Max-product inference (also called MAP inference) enables queries about the mode of a distribution.

As all other inference tasks, this is, in general, intractable.

For sparsely connected graphical models, we can use a variant of VE

- use factor maximisation instead of factor marginalisation
- traverse intermediate factors in reverse order solving a recursive argmax problem (the ‘traceback’) — aka ‘decoding’

Max-prod VE gives us a solution to MAP queries whose computational feasibility depends on the size of the largest intermediate factor constructed in VE.

What Next?

WC3: exam-like exercises.

Next week: midterm.

Next module: approximate inference.

References I

[1] Daphne Koller and Nir Friedman. *Probabilistic graphical models: principles and techniques*. MIT press, 2009.